

Digital Twins in Building Renovation

A Literature-Based Investigation into Use Cases, Value Creation, and Economic Evaluation Methods

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Callista Raschauer

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Kurzfassung

Angetrieben durch den alternden Gebäudebestand in Europa und Nachhaltigkeitsziele der EU hat die Gebäuderenovierung zunehmend an Bedeutung gewonnen, wobei die Betriebs- und Instandhaltungsphase hierbei ein entscheidender Faktor für dauerhafte Energie- und CO₂-Einsparungen ist. Diese Arbeit untersucht, wie Digitale Zwillinge (DTs) bei der Gebäudesanierung eingesetzt werden können und wie ihre Vorteile bewertet und quantifiziert werden können. Zunächst werden die Funktionalitäten von DTs in den drei Phasen Planung, Zustandsbewertung und -überwachung sowie Betrieb und Gebäudeverwaltung bewertet. Insbesondere die DT-gestützte vorausschauende Instandhaltung erweist sich als wichtiger Schlüsselfaktor, da sie die Echtzeit-Anomalieerkennung mit einer zustandsbasierten Planung kombiniert, um ungeplante Ausfallzeiten zu reduzieren und vermeiden und somit die Betriebskontinuität, Kosteneffizienz und Nachhaltigkeit sichergestellt wird. Um den Wert eines DT zu verdeutlichen, werden dessen Vorteile in Kombination mit etablierten wirtschaftlichen Modellen für eine Bewertung herangezogen, um den Nutzen mit Kosten zu vergleichen und Investitionsentscheidungen zu ermöglichen. Die Lebenszykluskostenanalyse macht unter Einbeziehung von DT-Daten statische Schätzungen zu adaptiven Prognosen. “What-if”- und Sensitivitätsanalysen innerhalb des DT decken frühzeitig die wichtigsten Kostentreiber wie Materialauswahl und Degradationspfade auf, was frühere und gezieltere Maßnahmen ermöglicht. In der Praxis unterstützen DT zunehmend die Lebenszykluskostenanalyse selbst, da sie Sensor- und Betriebsdaten kombinieren, um präzisere, kontinuierlich aktualisierte und entscheidungsrelevante Kosteneinblicke zu liefern. Das Return on Investment Modell wird bereits häufig angewendet, da es vertiefte Einblicke in finanzielle Abwägungen bietet und eine detaillierte Aufstellung der Kosteneinsparungen und vermiedenen Kosten erfordert. Verschiedene Bereiche wie die Hightech-Fertigung oder erneuerbare Energien haben durch die Einführung von DT einen deutlichen Anstieg des Return on Investments verzeichnet. Insgesamt kann der Return on Investment für DT zwar methodisch komplex sein, wird jedoch zunehmend robuster, wenn es mit einer Lebenszyklusanalyse und einer Multi-Szenario-Simulation kombiniert wird, die sowohl den kurz- als auch den langfristigen Wert erfasst. Dennoch stehen Hindernisse wie begrenzte Langzeitdaten, Interoperabilitätsprobleme und Lücken in der Governance oder Ausbildung einer routinemäßigen Einführung im Wege. Durch die Verknüpfung funktionaler Anwendungsfälle mit Quantifizierungsmethoden bietet diese Arbeit Wege zur effektiven Bewertung von DT-Investitionen.

Abstract

Driven by Europe’s aging building stock and EU sustainability targets, renovation has become increasingly important, especially with the operations and maintenance phase being a critical lever for lasting energy and carbon reductions. This thesis investigates how Digital Twins (DTs) can be operationalized in building renovation and how their benefits can be assessed and quantified. First, DT capabilities are assessed across the three phases: planning, condition assessment & monitoring, and operations & facility management, highlighting the different functionalities like real-time anomaly detection and predictive maintenance. In particular, DT-enabled predictive maintenance emerges as a primary value driver by combining real-time anomaly detection with condition-based scheduling to cut unplanned downtime, avoid failures, and support building operations and thus ensuring operational continuity, cost efficiency, and sustainability. Second, to clarify the value of a DT, their benefits in combination with established economical frameworks are used for an assessment to compare benefits with costs and make investment decisions. Firstly, Life Cycle Cost Analysis integrating DT data turns static estimates into adaptive forecasts. “What-if” and sensitivity analyses within the DT expose dominant cost drivers early, such as material choices and degradation paths, enabling more timely, targeted interventions. In practice, DTs increasingly support the Life Cycle Cost Analysis itself, as they combine sensor and operational data to deliver more precise, continuously updated, decision-ready cost insights. Secondly, Return on Investment is already widely applied as it offers actionable insight into financial trade-offs and requires a detailed accounting of cost savings and avoided costs. Various domains like high-tech manufacturing or renewable energy have shown significant Return on Investment increases from DT adoption. Overall, while Return on Investment for DT can be methodologically complex and context-dependent, it is increasingly robust when integrated with life cycle analysis and multi-scenario simulation, capturing both short- and long-term value. Nonetheless, barriers such as limited longitudinal evidence, interoperability issues, and governance or training gaps hinder routine adoption. By connecting functional use cases with quantification methods, this thesis provides pathways for effectively evaluating and prioritizing DT investments.

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Introduction

1.1 Motivation

The renovation of existing buildings has become an increasingly important topic in recent years, driven both by the pressures of an aging building stock and by the ambitious sustainability goals of the European Union (EU), which include achieving climate neutrality by 2050, significantly improving energy efficiency and reducing greenhouse gas emissions [47, 44]. According to the European Commission, 85% of EU buildings were built before 2000, and 75% of these have poor energy performance [45]. The urgency is further underscored by the fact that over 90% of the world's current building stock is expected to still be in use in 2050, meaning that much of today's inefficient stock will continue to impact energy consumption and emissions for decades [88].

Given these figures, the need for effective renovation strategies is particularly pronounced in regions like Europe, where the building sector accounts for a significant share of energy use and carbon emissions. In response, governments and international bodies have introduced policies and targets to accelerate renovation. For example, the EU's climate strategy and regulations call for dramatically improving the energy performance of buildings [47, 44]. In this context, the operations and maintenance phase of a building's life cycle stands out as one of the most critical periods in terms of energy consumption and sustainability, in which Digital Twins (DT) can deliver ongoing value and support achieving the climate goals of the EU [54]. By enabling continuous monitoring, predictive maintenance, and performance optimization, DTs help ensure that renovated buildings operate safely, efficiently, and sustainably over time [108, 9].

1.2 Problem

Undertaking renovation projects remains a complex challenge, as, unlike new construction, renovation involves working with existing structures that often hide unforeseen issues,

making project planning and execution difficult [104, 62]. The complexity of coordinating multiple trades in old buildings and dealing with surprises on-site can lead to higher costs and schedule overruns. Uncertainty is an inherent trait in renovation as project teams might not fully know the building’s condition until work is underway, and performance outcomes can be hard to predict [73, 109]. A further hurdle is the frequent lack of digital documentation for older buildings. Many legacy buildings have incomplete or out-of-date plans and no Digital Models, forcing renovation teams to spend considerable effort on surveys, measurements, and planning for later maintenance [153, 109]. In this context, digital tools offer promise: a DT could significantly improve information availability and decision-making. By creating a detailed Digital Model of an existing building and integrating real-time data from sensors or building management systems, a DT can allow stakeholders to visualize the current state, simulate renovation options, monitor progress in a virtual environment, and later on support the maintenance phase of the building [60]. Yet despite this potential, measuring the economic value of using DT technology in renovation projects is not straightforward. Stakeholders such as building owners and contractors often question whether the benefits justify the investment. The concept of a DT is relatively new in the renovation industry, and there is no consensus on how to quantify its return on investment or overall economic impact [109, 25]. This lack of clarity on value – combined with the upfront costs of creating and maintaining a DT – means that adoption in practice has been cautious. In summary, the renovation sector faces the pressing challenge of determining the added value of a Digital Twin and establishing methods to demonstrate and measure it in economic terms.

1.3 Primary Research and Supporting Questions

To address this gap, this thesis is guided by the primary research question:

“What are the existing use cases of Digital Twins in building renovation, and how can their economic value be assessed and quantified?”

This overarching question is supported by several more specific questions that will structure the inquiry:

- **Use Cases:** What are the primary use cases for Digital Twin technology in building renovation projects? How and where in the renovation process can Digital Twins be applied – especially in the operation and maintenance phase – and what functions do these use cases serve?

Those questions assess where and how Digital Twins create functional value throughout the renovation life cycle, particularly in the operation and maintenance phase, thereby supporting a deeper understanding of their potential for later economic quantification.

- **Created Value:** How can the value generated by Digital Twins be assessed?

This question examines what benefits Digital Twins provide through their different functionalities.

- **Evaluation Methods:** What methods or metrics can be used to evaluate and quantify the economic value of Digital Twins in renovation projects?

This includes exploring appropriate economic evaluation frameworks that capture the financial dimensions and benefits associated with implementing a Digital Twin, and discussing applications to illustrate practical outcomes.

1.4 Objectives of the Thesis

Thus, the objectives of this thesis can be described as follows:

- Review the literature on Digital Twins within the building sector, clarifying their definition, underlying technologies, and documented benefits, particularly in the context of building renovation.
- Identify and categorize the use cases of Digital Twin functionalities in renovation projects, and map them to distinct renovation phases as well as to the specific functions they serve.
- Analyze the types of value generated by these applications, including operational efficiencies, cost savings, sustainability impacts, and enhanced decision-making throughout the renovation life cycle.
- Explore and propose methods for quantifying economic value and discuss suitable frameworks to enable an assessment of Digital Twin investments in renovation.

1.5 Scope and Limitations

The scope of this research is deliberately focused and has certain limitations. Firstly, the study focuses on the renovation of buildings rather than new construction. While many principles of DTs overlap with new building projects, renovation brings distinct challenges and is therefore treated as a separate context. Secondly, the work is primarily a theoretical and literature-based investigation, meaning it synthesizes findings from academic research, industry reports, and documented examples, but it does not include a full empirical case study or the development of a new DT tool. Real-world cases or examples may be referenced to illustrate points, but the analysis relies on previously published data and qualitative insights rather than collecting primary data from a specific project. This approach offers a broad overview of use cases and value considerations, whilst meaning that certain technical details or organization-specific factors fall outside this paper's scope. In addition, the economic evaluation will be discussed in general terms rather than calculated for one particular project. These limitations imply that

while the thesis aims to present generalizable insights and recommendations, it might not capture every aspect of the complexity of real-world DT implementations.

1.6 Outline of the Thesis

Finally, the structure of the thesis is organized to systematically address the research questions and objectives. Chapter 2 will review the relevant literature and background, covering topics such as the current state of building renovation needs, the concept and technology of DTs and related digital tools, and prior research on the use of DTs in the building industry. This chapter establishes the foundation and context for the subsequent analysis. Chapter 3 will outline the research design and methodology, detailing the approach taken to identify, analyze, and categorize use cases, as well as to explore economic evaluation methods. Chapter 4 will focus on the identification of DT use cases in renovation projects. It will describe and categorize the ways DTs can be applied across the renovation life cycle, drawing on examples and studies to illustrate the use case. Chapter 5 will examine the economic value of these DT applications. Additionally, the benefits observed or claimed for each use case are analyzed, and potential methods to quantify those benefits are discussed. Lastly, Chapter 6 will synthesize the findings, providing a discussion on how the use cases and value quantification come together and will conclude the thesis by summarizing the key insights, discussing limitations, and offering recommendations for future work. Through this structured approach, the thesis aims to contribute to a clearer understanding of where DTs fit into renovation processes and how their economic value can be assessed, thereby supporting more informed decision-making in the pursuit of sustainable, efficient building upgrades.

Fundamentals

This chapter aims to provide an overview of the most relevant concepts related to DTs and their application within the context of the renovation industry. By outlining the foundational principles of DTs, examining the unique characteristics of the renovation sector, and exploring how DT technologies intersect with renovation processes, this chapter seeks to establish a comprehensive basis for the discussions and analyses that will follow in the later parts.

2.1 Digital Twins

2.1.1 History of Digital Twins

The term “Digital Twin” (DT) has become a central part of the industry’s digital revolution but, conceptually, has been around for many decades. One of the earliest practical expressions of the “Twin” concept can be found in the Apollo missions of the 1960s, when National Aeronautics and Space Administration (NASA) engineers built an exact physical system back on Earth to model and track the behavior of space missions in flight [58]. These physical duplicates allowed mission control to compare live data to react to problems that had arisen, encapsulating the principle of keeping a real-time representation of a physical entity in synchronism [121]. While the concept of mirrored systems had been explored earlier, the concept of the DT can be traced back to October 2002, when Michael Grieves first presented the model, which was back then unnamed, at a Society of Manufacturing Engineers (SME) conference as part of a broader vision for Product Life Cycle Management (PLM). Grieves [53] defined three core elements to his model: a physical space, a virtual space, and the data that serves to connect them. The term itself originated in 2010 with NASA’s roadmap based on the concepts developed during the Apollo era. In this context, a DT was described as a mirroring digital representation of a physical system, with sensor data replicating

the behavior to aid in decision-making and was originally implemented widely in the aerospace industry to monitor the health of aircrafts [136]. By continuously integrating sensor telemetry, these virtual counterparts furnish an unbroken, real-time “window” into aircraft operations, which empowers operators with unprecedented system visibility and rapid fault response [80].

In these early stages, DTs primarily focused on mirroring physical assets. This was closely aligned with the established practice of using Simulation Models, which replicate physical systems in virtual environments to analyze behavior and predict outcomes. However, unlike DTs, Simulation Models primarily rely on historical data and lack real-time synchronization with their physical counterparts. Thus, while they formed a critical foundation for the development of DTs, they lack the dynamic, bidirectional data integration that defines modern DT systems [157].

From the mid-2010s, interest in DTs surged in a more general context, driven by the advent of enabling technologies like Internet of Things (IoT), cloud computing, and Artificial Intelligence (AI). This was particularly evident in manufacturing under the umbrella of Industry 4.0, as well as in construction and facility management sectors aiming to improve operational efficiency and decision-making [121]. In parallel, the concept of Cyber-Physical Systems (CPS) emerged as a complementary but distinct technology paradigm. CPS integrates digital and physical components with embedded sensors, communication, and control technologies, enabling real-time interaction. Unlike DTs, which maintain tightly coupled one-to-one virtual representations of physical counterparts, CPS often facilitate one-to-many mappings and emphasize autonomous operation and control, deeply rooted in scientific and theoretical foundations [140].

As DT technology matured, its application scope expanded significantly beyond physical asset representation. Initially developed to mirror tangible objects, DTs were increasingly recognized as valuable enterprise assets. Dietz and Pernul [39] emphasize that DTs now manage both tangible and intangible assets, ranging “from turbines to services”, and provide value by integrating separated data streams across an asset’s life cycle. They further argue that DTs contribute to value creation by promoting system integration and supporting business transformations across company boundaries through a system-of-systems approach [39]. This enterprise perspective is reinforced by Mihai et al. [94], who describe DTs as evolving “*far beyond traditional computer-based simulations*” into platforms that enable seamless monitoring, analysis, and prediction [94]. They also note that this evolution has been further accelerated by external pressures, especially due to the COVID-19 pandemic, which prompted many companies to invest in DTs as part of broader digital transformation strategies to increase resilience and adapt to market disruptions [94].

2.1.2 Terminology and Characterization

The concept of a DT has become a central part of today’s digital transformation efforts across various industries. In general, a DT is a digital replica of a physical product,

process, or system, kept in synchronization by continuously updated data from sensors and IoT devices in real-time [42]. Through such a connection, one can monitor, simulate, and analyze the physical asset's status and performance in a precise manner. Simply put, a DT is a virtual representation of a physical object, system, or process that remains in synchronization with the actual physical entity through live sensor and IoT device data [67].

Despite its growing popularity, the term “Digital Twin” is still misused or, in some cases, conflated with other related technologies. VanDerHorn and Mahadevan [150] identify this ambiguity as a barrier to implementation. The two compared 46 definitions and suggested a consolidated definition: “*A virtual representation of a physical system (and its associated environment and processes) that is updated through the exchange of information between the physical and virtual systems*” [150]. Singh et al. [136] also explain the difference between a DT and other similar concepts. A Digital Model is a static representation; a digital shadow has one-way data flow, and a DT has unique, real-time, two-way synchronization with the physical counterpart. This concept will be discussed in the later section 2.1.4 discussing the “Level of Data Integration”. The real-time feedback loop resulting from this is what provides DTs with predictive and diagnostic analysis [136]. In addition to these core features, Trauer et al. [147] stress that a DT should extend across the entire product life cycle. This means it is not limited to a single phase but is continuously linked to the physical system from design and production to operation, maintenance, and disposal. At each stage, the twin enables real-time feedback, predictive insights, and data-driven decision-making, ensuring long-term value and adaptability [147]. Semeraro et al. [127] adopted a similar approach by emphasizing the integration of life cycle phases in their conceptualization of a DT. They conducted a systematic literature review of 150 peer-reviewed publications, using text mining techniques and formal concept analysis to extract key themes and structural patterns. Based on this, they defined a DT as a dynamic ensemble of adaptive virtual models that emulate the behavior of a physical system across its entire life cycle. Continuously updated with real-time data, the DT replicates the physical system to predict failures and opportunities for change, and to prescribe real-time actions for optimizing and/or mitigating unexpected events by observing and evaluating the system's operational profile [127]. Building on these perspectives, and in response to the ongoing ambiguity in terminology, the following definition is proposed to reflect a comprehensive, life cycle-oriented, and application-ready view of DTs that will be used in this paper:

“A real-time synchronized virtual model of a physical system that spans its entire life cycle, enabling prediction, optimization, and adaptive decision-making through continuous data exchange.”

The adoption of a standardized terminology and reference architectures is not only a theoretical concern; it is a critical prerequisite for practical deployment. As Botin et al. [22] emphasize, the lack of a universal and standardized definition of DTs prevents broader implementation across domains, making interoperability, integration, and scalability considerably more challenging in real-world contexts. In order to ensure that future

developments are transferable across domains and in line with societal and technological readiness, the authors point out that closing these standardization gaps will ultimately be necessary to achieve a widespread and mature implementation of DTs [22]. While the provided definition for this paper establishes a conceptual foundation, this review does not aim to resolve definitional ambiguities. Instead, it adopts a functional perspective, focusing on commonly associated functionalities of DTs. This offers a practical basis for further analysis, which will be discussed in the following chapters.

2.1.3 Structural Foundations of Digital Twins

Similar to the ongoing debate around the definition of a DT, its structural composition also lacks a universally accepted standard. As highlighted by Semeraro et al. [127], various studies propose divergent architectural models shaped by application context, technological maturity, and system complexity. This variation reflects not just differences in implementation strategies but also in how papers conceptualize the underlying components of a DT [127].

The system architecture of a DT defines its high-level structure, behavior, and views, encompassing both the digital and physical elements. It describes how various subsystems and components, such as sensing, modeling, and control functions, are organized and interact to achieve the system's objectives. In contrast, the components themselves represent the technical building blocks required to implement these functionalities [143].

While many foundational frameworks [22, 53] mainly define DTs by using three principal components – the physical reality, the virtual representation, and the information interconnection – more advanced models [142] expand this structure. One such example is the five-dimension architecture proposed by Tao et al. [142], which adds dedicated layers for services and DT data to support more intelligent operations and enable seamless data fusion between physical and virtual domains [142]. Similarly, Ponomarev et al. [115] propose a five-layer DT architecture tailored for complex technological processes. Their model includes a cyber-physical control layer, a primary data processing layer, a distributed computing and storage layer, a modeling and algorithmic layer, and a visualization/user interface layer. Each layer functions as a semi-autonomous subsystem, allowing for modular development and multi-user access. By integrating technologies such as cloud computing, big data analytics, and distributed control, this architecture emphasizes scalability, responsiveness, and interoperability – attributes essential for modern industrial environments [115]. These additional dimensions facilitate high-fidelity modeling, enriched analytics, and dynamic interaction capabilities, particularly beneficial for complex applications such as prognostics and health management [127]. Despite these variations, a simplified structure generally centers the following three fundamental components [141, 150, 165]:

- **Physical Reality** refers to the *actual system being modeled*, its environment, and the processes it undergoes. This includes machinery, structures, natural systems, or

any entity whose state can be observed and measured. The physical environment, referring to external conditions affecting the system, and the dynamic processes like wear, loading, or environmental interactions are integral to this component [150].

- **Virtual Representation** serves as the *digital mirror of the physical product* throughout its life cycle. It includes not only geometric models but also simulates and encodes material properties, mechanical behavior, health monitoring, and system rules. It is continuously updated through two-way data flows between physical and digital spaces and enables real-time monitoring, prediction, and control by integrating both observed data and derived information [141].
- **Information Interconnection** acts as the *communication bridge between the real and virtual environments*. Data collected from manufacturing systems, sensors, and databases undergo processing steps – acquisition, cleaning, analysis, and fusion – before being mapped into the Digital Model. This exchange may occur in real-time through integrated controllers or in delayed modes via temporary storage systems, supporting consistent system awareness, operational insight, and informed responses [165].

Together, these components make the DT a dynamic and evolving model that supports activities such as predictive maintenance, performance optimization, and scenario analysis across diverse industries [150].

2.1.4 Level of Data Integration and Flow

Data flow is a core differentiating factor in how digital representations of physical systems are classified and understood. According to Kritzinger et al. [76], the concepts of Digital Model, Digital Shadow, and DT each reflect a distinct level of data integration and interaction with the physical asset. While these terms are often used interchangeably, they vary significantly in their synchronization and communication capabilities – from no automated connection in Digital Models, to unidirectional data flow in Digital Shadows, and finally to full bidirectional communication in DTs, where the digital entity can both observe and influence the physical counterpart [76].

Digital Models: Static Representations

Digital Models (DM) are foundational representations used extensively in engineering and manufacturing for tasks such as design, documentation, and offline simulation [76]. However, these models are typically static and isolated from real-time system behavior. As Wright and Davidson [158] explain, such models can capture the state of a system at a specific point in time but do not update automatically with incoming data from the physical object. The lack of continuous data exchange means they are limited in capturing dynamic changes or evolving system states [158]. Thus, DMs offer a one-directional or absent data flow as well as a static structure. Furthermore, they are updated manually

or in batches and are not inherently linked to the live conditions of the physical system. As such, while useful for early-phase validation and design, DMs fall short in applications requiring adaptive behavior or real-time responsiveness [76, 158].

Digital Shadows: Automated Unidirectional Flow

Digital Shadows (DS) represent an intermediate stage between static DMs and fully interactive DTs [76]. Unlike a DM, which lacks automated data integration, a DS is characterized by an automated, unidirectional flow of data from the physical object to its digital counterpart [16]. This means that while the digital representation receives continuous updates, it cannot influence or send feedback to the physical system [76, 16].

In manufacturing settings, DS play a vital role in generating real-time, object-specific data mappings, enabling situational awareness without initiating physical system adjustments [16]. Because of their strictly one-way data exchange, they are particularly suited for monitoring and diagnostic tasks, where real-time control or optimization is not required [76, 16]. Ladj et al. [77] expand on this by presenting a DS as an enabler for structured data analytics, integrating sensor data and domain knowledge into context-aware platforms. These platforms support incident detection, semantic enrichment, and performance monitoring through business rules and expert input, enhancing data value and informing future DT applications [77].

Digital Twins: Bidirectional Synchronization and Interaction

DTs are the most advanced form of digital representation, characterized by their bidirectional synchronization with the physical entity, which enables continuous feedback and dynamic alignment between virtual and physical systems [52]. This bidirectional communication allows the digital counterpart not only to receive data from the physical object but also to send back decisions, control actions, or optimization parameters, thereby creating a closed-loop integration [82].

Through this dynamic interaction, DTs can evolve in parallel with the physical asset, adapting to both environmental conditions and system-internal changes in real time. This adaptability makes them capable of autonomous responses, where real-time monitoring and simulation enable DTs to support on-the-fly decision-making and adjustment [48].

DTs are designed to consolidate real-time sensor data, historical logs, and contextual domain knowledge, building a comprehensive virtual model of the system [82]. This virtual model is used for simulating future behavior, forecasting operational trends, and exploring possible outcomes before enacting changes in the physical domain [48]. Within industrial settings, DTs are widely applied for predictive maintenance, fault detection, and process optimization, offering a tool for proactive system management [82]. They play a critical role in smart manufacturing, improving uptime and reliability while enabling data-driven decisions throughout the entire system life cycle [52].

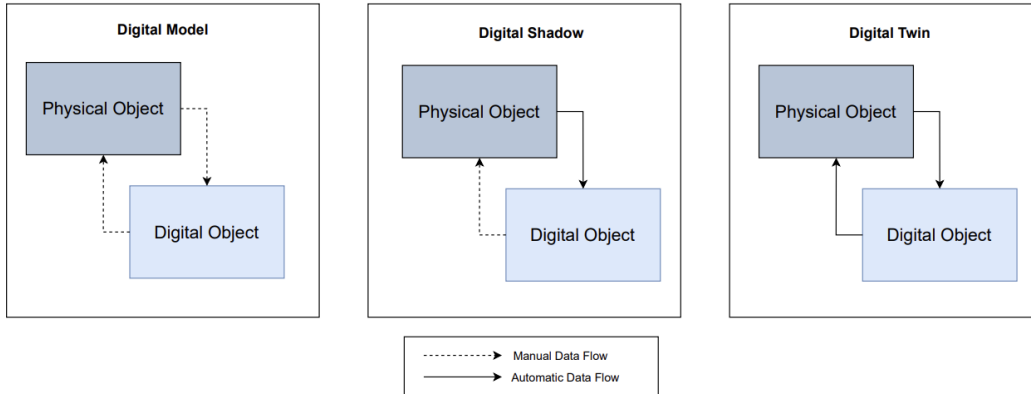


Figure 2.1: Types of DT based on level of integration adopted from [136]).

Information Flow as the Defining Characteristic

The essential distinction between these constructs lies in their information flow mechanisms, summarized in Table 2.1.

Table 2.1: Overview of information flow mechanisms across digital representation types. In adaptation to [76].

Concept	Flow Type	Flow Direction	Real-time Capabilities
Digital Model	Manual	None	No
Digital Shadow	Automated	Physical $\rightarrow$ Digital	Yes (monitoring only)
Digital Twin	Automated + Interactive	Physical $\leftrightarrow$ <i>Digital</i>	Yes (monitoring + control)

As Wright and Davidson [158] emphasize, distinguishing between DMs, DSs, and DTs is essential to avoid mislabeling and overhyping technologies. Misusing the term “Digital Twin” for static or one-way systems risks eroding trust and obscuring the true potential of bidirectional, real-time integration that defines actual DTs, which – as Kritzinger et al. [76] affirm – serve as a cornerstone for enabling intelligent decision-making and optimized control in Industry 4.0 environments [158, 76].

2.1.5 Application across industries

The versatility of the DT technology has resulted in its applicability across various industries, each utilizing its fundamental properties such as real-time synchronizing, data-driven modeling, and simulation potential to maximize efficiency, enhance reliability, and innovation [10]. Below listed are some of the most common applications of DTs:

- **Manufacturing:** DTs are widely used for product life cycle management, predictive maintenance, and real-time process optimization. By integrating continuous machine and sensor data into virtual models, manufacturers can anticipate equipment health, minimize downtime, and enhance quality control [100].
- **Medical and Healthcare Devices:** Another commonly used application of DTs are employed in personalized medicine to simulate physiological conditions. Patient-specific digital replicas of body parts assist clinicians in predicting outcomes and tailoring treatment plans before real-world application [138].
- **Energy Systems and Electric Vehicles:** In the context of the energy domain, particularly for Electric Vehicles battery systems, DTs facilitate life cycle monitoring, safety diagnostics, and predictive performance analysis. Through the application of AI enhanced models, operational safety and guide battery maintenance strategies can be improved [99].
- **Agriculture:** DTs in agriculture integrate IoT-enabled sensors, remote sensing, and data-driven simulations to build high-resolution, near-real-time virtual models of fields, crops, and livestock. By enabling continuous monitoring and “what-if” scenario analyses, such as simulating irrigation schedules or predicting pest outbreaks, these systems support early anomaly detection and precision interventions, thereby improving yield optimization and sustainable resource management [117].
- **Construction and Facilities Management:** In the building sector, DTs support space utilization, energy retrofitting, and planned maintenance. When integrated with Building Information Modeling and IoT, they allow continuous performance tracking and data-driven renovation strategies [109]. This application will be explored in depth in later chapters.

This widespread adoption of DTs across diverse sectors highlights their adaptability as a foundational digital infrastructure. As industries and governments increasingly rely on DTs for tasks like urban planning, sustainability tracking, and intelligent systems management, their role in enhancing automation, resilience, and long-term efficiency becomes even more critical [22, 48].

2.2 Renovation Industry

The Role of Renovation

Over the past few years, building renovation has been gaining more and more importance across Europe due to an aging post-World War II building stock, rising energy costs, and growing sustainability concerns. Many spend most part of their life inside buildings, which deteriorate over time, meaning ensuring healthy, energy-efficient indoor environments is a vital aspect not only for occupant well-being but also for reducing fossil fuel use and CO₂ emissions [68]. With the building sector being responsible for nearly one third of the greenhouse gas emissions and a significant share of carbon emissions, the importance of energy reduction and thus renovation increases every day [97].

Positioning Renovation Within the Building Life Cycle

Renovation is not a stand-alone task but an integral and recurring part of the building life cycle. According to the IG Lebenszyklus Bau [59], a building's life cycle is structured into six main phases: strategy, initiation, planning, execution, usage, and transformation. Renovation activities primarily occur in the latter two: the usage phase (e.g., maintenance and adaptation) and the transformation phase (e.g., repurposing, partial deconstruction, or redevelopment). Key renovation activities such as refurbishment (Instandsetzung) and functional adaptation (Umnutzung) aim to ensure continued performance, extend the building's useful life, and respond to shifting user needs or regulatory demands [59].

Moreover, Nielsen et al. [103] state that early-phase planning is vital to sustainable building renovation, as setting clear goals and criteria guides the direction of the entire process. Decision support tools that facilitate this step help stakeholders integrate sustainability considerations from the outset, fostering informed discussions and aligning objectives, which ultimately adds value and leads to more effective renovation outcomes [103].

The Scale of the Challenge

According to the European Commission, 85% of EU buildings were built before 2000, and 75% of those have a poor energy performance. Furthermore, buildings consume about 40% of Europe's energy (mostly for heating and cooling), meaning inefficient buildings contribute heavily to emissions and energy waste [45]. Currently, only 1% of the total building floor space is being renovated yearly, which is significantly below the level that would be required to contain the problem [17]. The slow rate is alarming given that over 90% of the world's current building stock is anticipated to still be used up to the year 2050 [88].

Policy Initiatives: The European Green Deal

To solve such challenges, the European Green Deal [47], along with its Renovation Wave Strategy [44], have been established as strategic solutions for accelerating the

transformation of Europe’s building landscape. The Green Deal aims for an ambitious target of taking Europe towards becoming climate-neutral by 2050, where the building sector is critical for fulfilling the vision. Since buildings are among the biggest consumers of energy, the European Commission is keen on doubling the yearly renovation pace within the next ten years while making sure that renovations translate into greater energy and resource efficiency [47, 44].

Practical Constraints in the Renovation Industry

These regulatory pressures are pushing the renovation industry to modernize this aging inventory of buildings for greater efficiency and compliance with new performance requirements. Renovation work, nevertheless, is prone to inherent complexity and uncertainty. Unlike new buildings, old buildings tend to experience insufficient and poor documentation, building data is commonly lost or degraded over decades of operation, and maintenance histories are limited [163]. That lack of reliable data implies that the actual condition of buildings and systems is often unclear until demolition or inspection, raising the prospect of unexpected problems. Renovation work must also deal with limitations of the existing structure, resulting in high design complexity and complicated construction sequences [153].

Furthermore, retrofitting current buildings for sustainability faces critical technical challenges, based mainly on structural constraints and outdated systems. Older buildings often weren’t constructed to accommodate the extra load of integrated elements such as solar panels or green roofs. Necessary structural reinforcements for this are often very costly. Moreover, incorporating new technologies with obsolete electrical, plumbing, and heating, ventilation, and air conditioning systems is intricate, pricey, and needs careful planning to ensure that building operations are not affected [62].

Barriers and the Role of Innovation

Liao et al. [81] identified three key barriers to sustainable building renovation, consisting of financial constraints, policy barriers, and design obstacles. To solve these issues, they highlighted the importance of innovation and new technology [81]. Looking at all of these listed challenges and issues that arise in the context of the construction industry, it becomes clear that innovative approaches such as DTs are essential in breaking through current barriers [96].

2.3 The renovation process

Various frameworks describe the different phases of a building’s life cycle. Abd Rashid and Yusoff [1] propose a comprehensive Life Cycle Assessment based structure consisting of pre-use (raw material acquisition, manufacturing, transportation), construction, use (operation and maintenance), and end-of-life (demolition, recycling, disposal) phases, highlighting the use phase as the most environmentally impactful and emphasizing

the need for consistent data and standardized methods in building assessments [1]. In contrast, Ramesh et al. [119] outline a life cycle energy framework with three main phases: manufacturing (material production, transport, construction), use (operational energy), and demolition (teardown and recycling), followed by calculations of total Life Cycle Energy (LCE) and Life Cycle Assessment (LCA) for environmental impact [119].

This paper adopts the building life cycle definition presented in the earlier mentioned IG Lebenszyklus Bau [59] from 2.2, a professional guideline published by IG Lebenszyklus Bau, which is an Austrian industry consortium comprising real estate developers, planners, facility managers, and public sector stakeholders. According to them, building projects are structured into six interconnected life cycle phases: *Strategy, Initiation, Planning, Execution, Operation, and Deconstruction/Redesign*. In the *Strategy* phase, stakeholders define overarching project and service goals aligned with business and sustainability objectives (“Ableiten einer übergeordneten Objekt- und Servicestrategie”). The *Initiation* and *Planning* phases include demand analysis, feasibility studies, and life cycle cost assessments for renovation or redevelopment options (“Machbarkeitsstudie”, “Lebenszykluskostenberechnung”). During *Execution*, design concepts are implemented with quality assurance and sustainability monitoring integrated throughout (“Überwachung der Umsetzung der definierten Nachhaltigkeits- und Förderkriterien”). In the *Operation* phase, performance tracking, adaptive facility services, and post-occupancy evaluations ensure ongoing optimization (“Post Occupancy Evaluation”, “Optimierung aller Services entsprechend der Nachhaltigkeitskriterien”). Finally, the *Deconstruction* phase includes dismantling or reconfiguring the structure with attention to reuse and sustainable material flows [59].

For the purpose of this paper, the renovation process is viewed as encompassing three specific and interrelated activities – *Operations & facility Management, Condition Assessment and Monitoring, and Planning* – each embedded within the broader life cycle phases defined by IG Lebenszyklus Bau. The first, *Operations & Facility Management*, is situated within the Operation phase and involves the ongoing management and performance optimization of building systems. The second, *Condition Assessment and Monitoring*, is also part of the Operation phase but serves as a transitional activity that informs and prepares the groundwork for subsequent planning decisions. The third, *Planning*, aligns directly with the Planning phase of the life cycle model and focuses on developing renovation strategies based on assessed conditions, future requirements, and long-term goals.

Operations & Facility Management

The Operation and Maintenance (O&M) phase spans the longest period in a building’s life cycle, surpassing all earlier and later stages combined. As Obradovic et al. [106] emphasize, even in comparison to the construction and demolition phases, the O&M phase represents the most extended and cost-intensive period of building use. Proper and regular maintenance during this phase is critical not only for ensuring continued functionality and safety but also for maximizing service life and minimizing long-term costs [106].

Moreover, numerous studies have shown that the O&M phase contributes significantly to both the financial and environmental impacts of a building. Islam et al. [61] report that, for instance, the operational phase alone can account for up to 90% of total greenhouse gas emissions in a building's life cycle, depending on usage patterns, design efficiency, and system boundaries [61].

In this context, DTs have emerged as a valuable tool for enhancing condition monitoring and facility management throughout the O&M phase. By integrating sensor data and historical records, DTs can continuously track the geometric and operational state of building components, allowing for early detection of degradation and maintenance needs. These models allow asset managers to track changes over time, enabling predictive maintenance and preventing failures in systems like urban sewer networks. Beyond technical maintenance, DTs support broader facility management functions like indoor monitoring, oversight of mechanical, electrical, and plumbing systems, and user behavior analysis, using IoT, AI, and Augmented/Virtual Reality (AR/VR) to deliver real-time data that boosts efficiency and comfort [110].

Condition Assessment & Monitoring

Before detailed planning can begin, it is essential to understand the current condition of the building in order to select appropriate retrofit strategies. As Pombo et al. [113] state, assessing the state of key elements – such as the building envelope, systems, and usage patterns – is crucial for identifying effective energy-efficiency measures and ensuring that retrofit decisions are grounded in actual performance data [113]. In this context, Building Information Modeling (BIM), a method for digitally representing a building's physical and operational attributes, offers significant advantages for condition assessment and monitoring. Alavi et al. [4] highlight how BIM can consolidate diverse data sources, including inspection records, component age, and deterioration indicators, into a unified platform that supports consistent and accurate evaluations. By integrating BIM with decision-support tools and probabilistic models, stakeholders can monitor degradation trends, assess risk levels, and visualize building conditions in real time. This approach enables early detection of deficiencies and more informed planning, making condition assessment a dynamic and continuous process rather than a one-time task [4]. As Dejacco et al. [36] further emphasize, systematic condition assessment enhances asset knowledge and allows stakeholders to evaluate degradation levels, identify maintenance needs, and allocate resources more effectively across a building portfolio. It supports the development of a reliable understanding of physical conditions, enabling more targeted and cost-efficient decisions on repairs, replacements, and upgrades while minimizing the risk of unexpected failures and inefficiencies [36].

Recent advances in DT technology further enhance condition assessment by combining BIM-based data models with real-time sensor inputs and occupant feedback. As demonstrated by Hosamo et al. [56], this integration allows facility managers to evaluate the actual condition of building systems more accurately, detect faults early, and make in-

formed maintenance decisions within a unified BIM environment, significantly improving the quality and responsiveness of renovation planning [56].

Planning

Effective planning is critical in renovation projects, particularly those aiming to achieve high-performance standards such as Nearly Zero-Energy Buildings. As Konstantinou et al. [74] emphasize, the planning stage must accommodate increasingly complex and integrated solutions by combining passive and active systems, whilst also addressing stakeholder coordination and information flow across the supply chain [74]. This phase involves translating strategic and conceptual goals into detailed technical designs, supported by life cycle-oriented cost planning, risk assessments, and feasibility analyses. Collaborative processes are emphasized, ensuring all disciplines contribute to energy, comfort, and functionality targets through integrated digital tools like BIM. The Planning phase also includes setting procurement strategies, defining schedules, and ensuring regulatory compliance – all of which lay the groundwork for a coherent, cost-efficient, and future-proof renovation process [59].

In this context, BIM serves as a foundation for developing DTs, supporting the creation of semantically enriched 3D models that enhance stakeholder coordination and decision-making. As Yang et al. [160] highlight, these models integrate geometric and descriptive data and can be linked with AI, sensors, and mixed reality tools to simulate planning scenarios, identify potential clashes, and optimize resource use. This integration enables more precise, data-driven planning and reduces the risk of material waste and rework, improving the overall efficiency of renovation outcomes [160].

2.4 Digital Twins in the Context of Renovation Industry

The construction industry and, thus, as an essential part of it, the renovation sector is currently undergoing a transformative change through the adoption of advanced technologies, as the need arises for enhancing the efficiency, sustainability, as well as decision-making during the whole building life cycle [131, 118]. According to Rajala et al. [118], the renovation business has grown significantly, particularly in developed regions, due to aging building stock and stricter environmental regulations. In this context, technologies like BIMs and DTs have emerged as critical drivers of this shift [131].

According to Nguyen and Adhikari [101], DTs in the context of the construction industry are based on the technology of BIM, providing structured environments, design documentation, and life cycle modeling. DT systems enhance the capabilities of BIM by allowing for real-time tracking, predictive maintenance, as well as operational optimization through IoT, sensor networks, and machine learning [101]. Thus DTs and BIMs and their functionalities are gaining more importance and popularity as a possible solution since the construction sector faces pressure to meet the prior mentioned stricter environmental targets [20].

Building Information Modeling (BIM)

As mentioned earlier, Building Information Modeling (BIM) refers to a digital methodology that facilitates the creation of a comprehensive and data-rich virtual representation of a building throughout its life cycle, from planning and design, through construction, to operation and maintenance [144]. BIM plays a central role in improving productivity and reducing fragmentation in the Architecture, Engineering, Construction, and Facility Management (AEC-FM) industry by enabling more effective communication, clash detection, design validation, and cost estimation [37]. BIM models are composed of objects that encapsulate both geometric and semantic (attribute) information, enabling stakeholders to visualize, simulate, and optimize real-world behavior during every project phase [3].

While BIM serves as a foundational component in the built environment, it does not inherently support the live synchronization, real-time analytics, or automated feedback loops that characterize a true DT. Only when BIM is enhanced with sensor data, IoT connectivity, and adaptive modeling does it begin to function dynamically as a DT, capable of reflecting and interacting with its physical counterpart in real time [20]. Moreover, recent literature emphasizes that while BIM provides a robust structure for data management and collaboration, it still suffers from limitations such as authorship, traceability, and transparency issues when managing dynamic information across the project life cycle [101].

The work by Deng et al. [37] provides a detailed taxonomy for understanding the progressive transition from static BIM to fully dynamic DTs in the built environment. Their five-level ladder model captures this evolution:

- **Level 1 – BIM:** Focuses on the static 3D representation of buildings, used for documentation, design, and visualization.
- **Level 2 – BIM-supported Simulations:** Adds simulations such as energy performance, lighting, and thermal analysis using data stored within BIM models.
- **Level 3 – BIM integrated with IoT:** Combines BIM with real-time data from IoT devices, allowing for monitoring of building status (e.g., temperature, occupancy).
- **Level 4 – BIM with Predictive Intelligence:** Incorporates machine learning and AI to generate forecasts and optimize performance, such as predictive maintenance or thermal comfort prediction.
- **Level 5 – Ideal DT:** Realizes a full feedback loop system in which sensor data, BIM models, simulations, and AI integrate to automate decisions and control systems, achieving real-time interaction with the physical asset.

This shows that BIM serves as a foundational step toward DTs, and through progressive integration of real-time data, IoT, machine learning, and predictive analytics, it can evolve

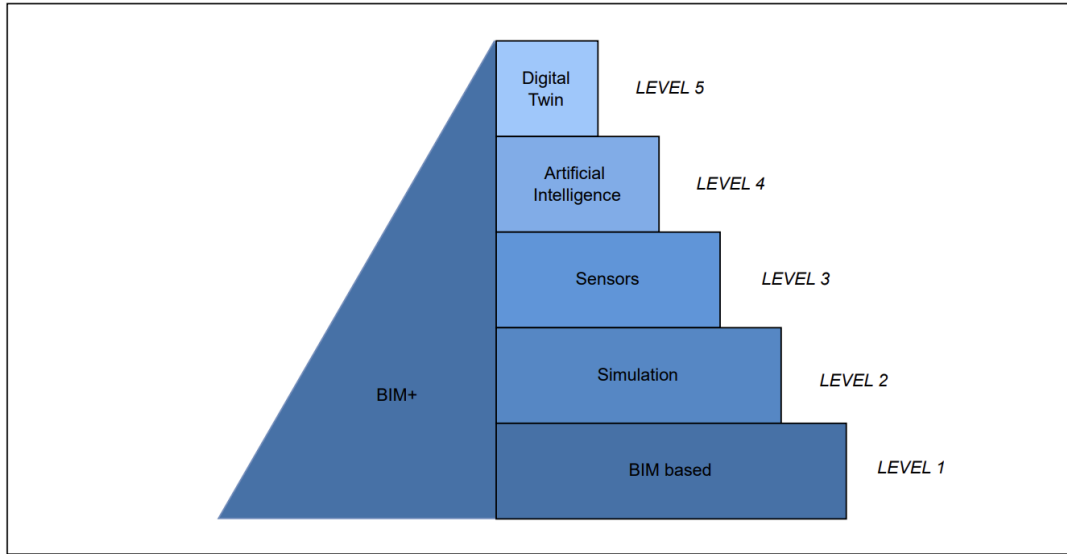


Figure 2.2: Taxonomy of the Evolution of BIM to DT in the built environment, adopted from Deng et al. [37].

into a dynamic and intelligent system capable of mirroring, analyzing, and optimizing the performance of physical assets throughout their life cycle [37].

However, growing use of terms like 4D or 5D BIM has sparked debate over the proper meaning of “dimensions” in BIM. As Koutamanis [75] argues, only spatial dimensions and time – used for construction sequencing – constitute true dimensions. Additions such as cost (5D) or sustainability (6D) are not intrinsic to the model but are derived from external data or calculations. These should be understood as supplementary information layers rather than structural dimensions. Misusing the term risks overstating BIM’s representational scope and obscuring its underlying data architecture [75]. Charef [28] even proposes an eighth dimension (8D) in the context of circular economy practices, specifically dedicated to managing the End-of-Life (EOL) phase of buildings. This 8D BIM integrates deconstruction information to simulate various EOL scenarios, aiming to maximize material recovery and sustainability outcomes [28].

Relevance of DTs

In renovation specifically, DTs are emerging as a key enabler of the EU’s Renovation Wave Strategy, supporting goals like energy efficiency, emissions reduction, and improved building performance through digital reconversion and smart readiness evaluation [78]. Renovation is different from a new building because it frequently starts with extremely uncertain, degraded information about the building’s condition. Documentation is missing, often outdated, and there is little information pertaining to structural systems or utilities due to decades of usage and modification [153].

Early applications show that DTs can enhance renovation processes through enabling accurate digital capture of a building’s current condition. Even when original documentation is missing, DTs support the creation of as-built models, allowing for better structural analysis and environmental monitoring. Through real-time data integration, DTs aid in early issue detection and strategy evaluation, ultimately improving decision-making and project outcomes [108]. During planning, DTs allow stakeholders to simulate various renovation strategies and evaluate their implications virtually before implementation [160].

In the construction execution phase and beyond, DTs support progress monitoring by enabling real-time tracking of components, performance metrics, and equipment interactions. These functionalities also extend to predictive maintenance and energy optimization during and after, ultimately enhancing decision-making and improving outcomes in cost, time, and quality [9].

Practical Applications in Renovation

From a functional perspective, several DT capabilities are particularly relevant for renovation projects. The “BIM4EEB” [43] toolkit, for instance, offers a modular and interoperable system that integrates BIM, IoT, and linked data technologies to support different renovation stages ranging from fast mapping of existing conditions to energy assessment, occupant feedback, and construction planning. Developed under the Horizon 2020 initiative (a research funding and innovation program), BIM4EEB is based on a web-based BIM Management System that complies with BIM Level 3 principles and enables real-time monitoring, collaboration, and visualization throughout the renovation process. By combining semantic modeling, sensor data integration, and user feedback tools, it facilitates more efficient workflows, improves decision-making, and supports the gradual evolution of BIM toward DT applications [33]. According to Bocullo et al. [17], at a larger scale, DT applications, such as those used in city block renovations, support simulation of renewable energy solutions, allowing planners to optimize heat and power systems before implementation and visualize outcomes in a user-friendly 3D environment [17]. When combined, these capabilities lower the risks associated with renovations, enhance collaboration, and produce better performance results in terms of energy, cost, and time [33, 17].

Challenges in adoption

Nevertheless, challenges from the adoption of DTs to renovation persist. A fundamental issue is the difficulty of obtaining reliable, real-time data from existing buildings that were not originally designed to support digital infrastructure. Retrofitting such structures with sensors and IoT systems often proves invasive, technically complex, and costly. Moreover, integrating DT platforms with existing BIM models or legacy asset management systems introduces further complications, particularly due to non-standardized data formats and the fragmented nature of historical documentation [73].

The effective deployment of predictive maintenance, real-time simulation, and decision-support functions also hinges on robust data governance and the availability of trained personnel capable of managing high-frequency, multi-source data streams. Interoperability remains another critical concern; the lack of standardized protocols across BIM and DT platforms limits seamless integration and scalability. Finally, the success of DT implementation in renovation contexts is often constrained by limited digital literacy among stakeholders and insufficient cross-disciplinary coordination, both of which are essential for sustaining complex digital ecosystems [63].

Despite these limitations, DTs are increasingly seen as a critical innovation for addressing systemic inefficiencies in renovation [108]. The following chapters will explore in detail how DT-related functionalities are applied across renovation phases and what specific value this provide in practice.

Research Design

3.1 Research Methodology

This thesis adopts a selective searching approach, focusing on identifying and analyzing relevant literature that explores how DT functionalities are applied and valued within the renovation industry, which is a critical phase in the building life cycle. The research is divided into two principal components. Firstly, it examines different use cases of DT-related technologies across key stages of the renovation process, like planning, condition assessment and monitoring, and operations and facility management, with a specific focus on the latter. Secondly, it investigates the value generated by these applications, both in qualitative terms and, where possible, quantitative terms, through the proposition of different economic models for value estimation.

In this study, a functional rather than strictly terminological approach is adopted, due to the inconsistent usage of the term “Digital Twin” in the built environment domain. Many sources refer to BIM-based models, often representing as-built DM, DS, or CPS under the same label. To ensure a comprehensive and inclusive analysis, this thesis does not limit the scope to publications that adhere to a narrow definition of a DT. Instead, it includes literature describing systems or processes which deliver functionalities commonly associated with DTs, such as real-time monitoring, for instance via IoT sensor integration, simulation and scenario analysis, or predictive maintenance enabled by predictive analytics. This functional lens enables a more accurate representation of how DT-like technologies are being implemented and discussed in the context of building renovation, even if different terminology is used.

3.2 Literature Selection Strategy

The literature for this study was selected through a systematic search of established academic databases, including *Scopus*, *Web of Science*, *IEEE Xplore*, and *Google Scholar*.

Search queries were constructed using Boolean operators (AND/OR) to combine key concepts and ensure comprehensive coverage. For example, queries included combinations such as “*Digital Twin*” AND “*building renovation*”, “*Digital Twin*” AND (*renovation* OR *retrofit* OR *refurbishment*), and “*Digital Twin*” AND (“*value quantification*” OR “*ROI*” OR “*LCCA*”).

These base terms were further expanded with logical OR operators to include synonyms and related keywords relevant to the topic. For instance, a search string in Scopus was formulated as: “**Digital Twin**” AND (“**renovation**” OR “**retrofit**”) to capture literature on DTs in renovation contexts. Additional queries targeted specific aspects by combining DT terminology with domain-specific keywords. Related concepts were also considered when constructing these queries. This Boolean search strategy ensured that differing terminology across disciplines was accounted for and that relevant publications were not overlooked due to vocabulary differences. Additionally, backward and forward citation tracking was applied, the reference lists of key papers were examined (backward tracing), and any newer works citing those key papers were identified (forward tracing), to uncover further relevant studies beyond the initial keyword searches.

To be included in the review, sources had to meet several selection criteria:

- *Relevance of Functionality*: The source should describe at least one functionality commonly associated with DTs.
- *Context of Renovation*: The study should be situated in the context of building renovation or the operation of existing buildings.
- *Clarity of Application*: The source should provide a clearly described application, methodology, or evaluation of the technology.

Regarding source type and timeliness, the review emphasized peer-reviewed academic papers as the primary body of literature, supplemented by selected industry reports from reputable organizations for additional practical insights. Furthermore, publications in English and generally those published within the last 10–12 years were considered, ensuring the research reflects contemporary developments. However, a few exceptions were made for foundational or frequently cited earlier works whose early contributions remain highly influential in the field.

3.3 Functional Framework

To structure the analysis, a framework of DT functionalities was developed based on patterns identified in the literature. These functionalities represent recurring technological capabilities that are typically associated with DTs in the context of buildings and infrastructure. Initial categories include, for instance, real-time monitoring, simulation, scenario analysis, and predictive maintenance.

This set of functionalities served as the foundation for analyzing both the use cases and the associated value dimensions. As the review progressed, the framework remained flexible and was refined or adjusted to reflect the variety and specifics of the examples found. Each use case extracted from the literature was classified according to the relevant functionality it demonstrates. This consistent categorization ensured comparability in the later analysis, making it possible to systematically contrast different cases and draw general insights despite variations in terminology or emphasis across sources.

Use Case Mapping

The first analytical part of the thesis focuses on identifying how DT-related functionalities are applied across the renovation process. The renovation process is divided into three main phases: planning, condition assessment and monitoring, and operations and facility management. For each phase, representative literature is reviewed to extract concrete use cases, which are then mapped to the functionality framework outlined above. From this broad survey, the analysis subsequently narrows its focus to use cases within the operations and facility management phase.

This mapping process enables a structured overview of how digital technologies are deployed in practice, even when described under varying terminologies. It also allows the identification of phase-specific patterns, showing, for example, whether simulation is predominantly used during planning or whether real-time monitoring is more common in the operational phase. The result is a consolidated view of current DT-related practices in renovation, grounded in documented examples.

Value Identification

The second analytical component of the thesis investigates the value that emerges from the application of DT functionalities in renovation contexts. The term “value” is understood broadly, encompassing both tangible benefits such as cost savings, time reduction, and energy efficiency, and intangible advantages such as improved decision-making, enhanced transparency, or regulatory compliance.

Value-related information is extracted directly from the reviewed literature, focusing on reported outcomes, evaluations, and assessments to provide a deeper understanding of how value is created from the adoption of DTs. Where available, quantitative data such as percentage improvements, financial figures, or performance metrics were also listed as examples. The values extracted from each use case are then consolidated into thematic categories. Building on this, the analysis further proposes economic frameworks suitable for quantifying the value generated by DTs and reviews existing implementations that integrate DTs within these frameworks.

3.4 Terminology and Scope

Given the inconsistent use of terminology in the field, this thesis adopts a pragmatic and functional approach. The term “DT” is used throughout the thesis as an umbrella concept referring to systems that demonstrate specific capabilities. This allows for the inclusion of studies that refer to digital shadows, BIM environments, or other systems, provided they exhibit functionalities consistent with the operational definition of DT used in this work.

This approach ensures both analytical clarity and inclusiveness, allowing the research to engage with the full range of relevant literature while maintaining a clear functional focus.

3.5 Limitations

The methodology is subject to several limitations. Firstly, the analysis relies exclusively on published literature, which may not fully capture the latest industry developments or undocumented practical implementations. Secondly, the functional approach may overlook technical distinctions between systems, as it emphasizes capabilities rather than architecture. Thirdly, the value identification is constrained by what authors report; in many cases, quantitative evaluations are limited or absent.

Despite these limitations, the chosen research design enables a systematic and focused examination of DT-related functionalities and their relevance for renovation. By combining a broad but targeted literature search with a functional analytical framework, the study provides an organized view of the current state of knowledge. Within the acknowledged constraints, this approach yields meaningful insights into how DT technologies are being applied in building renovation and what value they are perceived to bring, setting the stage for the deeper discussions and conclusions in the subsequent chapters.

Use Case of Digital Twins in Renovation

This chapter explores the application of DTs in building renovation, with a focus on their role in enhancing operations and maintenance through predictive maintenance strategies. It examines how DTs integrate real-time data and analytics to improve decision-making, reduce downtime, and support sustainable building management. Through key concepts, system architectures, and real-world examples, the chapter outlines the benefits, challenges, and practical implementations of predictive maintenance in facilities.

4.1 Operations & Management

The operation and maintenance (O&M) phase, which was earlier introduced in 2.3, is the longest and most costly stage in a facility's life cycle, yet it often receives limited focus. Many facilities still lack a clear asset management strategy and effective use of data. Maintaining long-term performance and safety increasingly requires a smart, digitally integrated approach, beginning from design through to O&M [54]. As Wong et al. [156] highlighted, despite the availability of enabling technologies such as BIM, reality capture and IoT sensors, their deployment in facility management remains sporadic and largely confined to isolated case studies, leaving significant gaps in areas like renovation, retrofitting and predictive maintenance [156].

This need becomes particularly critical in renovation projects, where operators must manage a hybrid environment of modernized technologies and aging infrastructure [27]. The integration of DTs during renovation provides not just real-time monitoring but also the ability to simulate future performance scenarios and optimize decision-making throughout the building's life cycle. Unlike static BIM models, DTs evolve in real-time by synchronizing with IoT-enabled assets, capturing live data such as temperature, vibration,

and energy use. This dynamic feedback loop enables predictive maintenance, allowing facility managers to detect system anomalies early, run diagnostics remotely, and even simulate the impact of delayed interventions, turning the building into an adaptive, self-aware environment rather than a static structure [63].

The next four functions described constitute the most prevalent DT applications in the operations and maintenance phase. (1) They provide real-time monitoring of environmental and system parameters, such as temperature, humidity, noise levels, occupancy, and energy consumption – through networks of IoT sensors. (2) By continuously processing live data streams, they can automatically diagnose faults and anomalies as soon as they arise. (3) They support performance simulation, enabling analyses that explore different energy-management strategies, lighting configurations, or emergency procedures before implementation. (4) They underpin predictive maintenance by using statistical and machine-learning models to forecast equipment needs and schedule interventions before failures occur [114].

Among the building systems managed in the O&M phase, Heating, Ventilation, and Air Conditioning (HVAC) systems are especially critical due to their impact on both energy performance and occupant well-being. Building systems refer to the interconnected technical subsystems within a building, such as HVAC, lighting, air quality monitoring, and energy management, that collectively ensure comfort, safety, and operational efficiency. These systems typically account for a large portion of a building's energy consumption and are prone to operational inefficiencies if not regularly maintained [130]. HVAC systems account for nearly half of the building sector's energy use, driven by their critical role in ensuring indoor air quality and thermal comfort [60]. By utilizing predicted indoor temperature trends, the system can schedule HVAC operations to minimize energy use during low-demand periods without compromising occupant comfort. Furthermore, by anticipating temperature fluctuations, the HVAC system can also adjust setpoints in advance, preventing temperature overshoots and reducing unnecessary equipment cycling [105].

These, besides many other capabilities, make DTs critical for optimizing energy consumption, extending equipment lifespan, and minimizing unplanned downtime through predictive maintenance [65, 102]. Furthermore, DTs support proactive decision-making by aggregating operational data across multiple building systems, including HVAC, lighting, and structural components [55].

As DT technologies evolve, they are increasingly being integrated with existing building management systems and facility information models, enhancing their utility for operations teams [55]. While challenges such as interoperability with legacy infrastructure and the cost of sensor deployment remain, the strategic advantages of DTs in the O&M phase are increasingly recognized as essential for sustainable, efficient, and intelligent facility operations [73, 98].

Facility Management

Facility Management is a key component of O&M, covering a wide range of building elements such as façades, flooring, air-conditioning systems, electrical components, and utility services. It also includes tracking and optimizing utility consumption to improve efficiency [162].

Comprehensive facility management brings together the planning, design, and oversight of a building’s physical assets to preempt operational failures and sustain continual service delivery [8]. Becker [15] defines facility management as follows: “*Facility management is responsible for coordinating all efforts related to planning, designing, and managing buildings and their systems, equipment, and furniture to enhance the organization’s ability to compete successfully in a rapidly changing world.*” By embedding systematic reviews of maintenance outcomes, cost efficiencies, and occupant satisfaction, it uncovers opportunities to enhance both environmental performance and user experience. The resulting insights are communicated to senior leadership, who then deploy targeted strategies to align facilities operations with broader organizational goals in productivity, profitability, and quality [8].

Moreover, facility management can be used for a structured comparison of current practices against recognized standards to identify where performance falls short, quantify the “distance” between actual and desired states, and formulate focused action plans for closing those gaps. This approach not only highlights priority areas for improvement but also underpins the continuous refinement of maintenance protocols, reporting mechanisms, and emergency procedures, ensuring that facility operations evolve in step with best-practice benchmarks [162].

DTs fundamentally transform facility management as their integration enables facility managers to move beyond static maintenance schedules toward a dynamic, data-driven approach that monitors asset health in real time, diagnoses emerging issues, and orchestrates targeted interventions before failures occur [114].

Applications during O&M

As Tuhaise et al. [148] highlighted in their review of DT technologies in construction, there are numerous applications of DTs [148], many of which can be effectively assigned to the Operations and Maintenance (O&M) phase based on IG Lebenszyklus Bau [59].

Predictive Maintenance

Among these, predictive maintenance is a prominent application utilizing real-time sensor data on vibration, temperature, and flow to forecast equipment failures before they occur, which will be discussed in depth in the next section [57, 30].

Real-time Monitoring and Anomaly Detection

In addition, real-time monitoring and anomaly detection are increasingly applied to monitor structural health (e.g., bridges and façades) and environmental conditions such as HVAC systems and tunnel ventilation. These approaches enable the immediate identification of operational deviations [116, 86, 159, 166, 72]. For example, in the paper by Khajavi et al. [72], a wireless sensor network was installed on the facade of an office building to continuously monitor light, temperature, and humidity levels. The collected data were processed and visualized in real-time to create a DT of the facade, enabling the detection of environmental changes and potential anomalies such as occlusions caused by nearby objects or people [72].

Performance Simulation and Analysis

Furthermore, energy and environmental management benefits from live building data, which can drive automated control systems and enable different scenario simulations to optimize HVAC and lighting efficiency [139, 18, 111]. In the paper by Borissova et al. [18], the authors instrument a single apartment with IoT sensors and integrate them via a software platform running on a Raspberry Pi (a single-board computer) to automate radiator valves based on both time schedules and window-open events. They then construct an identical 3-D BIM model of that apartment-its “DT”-and run a year’s worth of simulated heating loads on it, mirroring the real-world control rules. Finally, by directly comparing the measured energy consumption against the simulation results, they demonstrate a 47% reduction in total heating use after the DT-driven automation was applied [18].

Occupancy & Space-Utilization Analytics

Another growing field is occupancy and space-utilization analytics, which rely on technologies such as CO₂ sensors, smart cameras, and tracking systems like Radio Frequency Identification (RFID) or ultra-wideband (UWB). These tools are used to monitor how people move through and use spaces within a building. The resulting data helps forecast where and when heating, cooling, or lighting will be needed, thereby supporting more efficient space planning [126, 155, 91]. Seghezzi et al. [126] conducted a study to calibrate and assess a camera-based IoT sensor system for occupancy monitoring, as part of developing an occupancy-oriented DT for facility management. Their work involved test campaigns in a university office building to identify data collection issues, optimize sensor placement and orientation using BIM-supported post-occupancy evaluations, and evaluate data quality. The study demonstrated that with proper calibration and iterative testing, such systems can effectively monitor real-time space usage, enabling improved planning of cleaning services and space utilization [126].

Safety Risk Management

Moreover, safety risk management is supported by DTs that integrate sensor, vision, and positioning data to detect on-site hazards, such as fall risks on ladders or scaffolds, unsafe equipment use, or the early stages of fire spread. These systems can generate timely alerts and evacuation paths to enhance on-site safety [93, 83, 31]. In Cheng et al. [31], for instance, the authors built a BIM-based Intelligent Fire Prevention and Disaster Relief System by embedding Bluetooth-enabled smoke and temperature sensors (and RFID positioning tags) into a 3D DT of a completed building. As environmental and locational data stream continuously into the virtual model, computer-vision algorithms identify emerging fire or spatial hazards, and a graph-search engine (Dijkstra’s algorithm with risk-weighted edges) computes and dynamically updates the safest evacuation and rescue routes. Finally, turn-by-turn voice and arrow guidance is pushed to occupants’ and responders’ mobile devices, thus closing the loop between real-time sensing, simulation, and on-site action to enhance situational awareness and reduce response times [31].

Together, these diverse applications transform static BIM models into dynamic, data-rich platforms that enable proactive, efficient, and safe facility operations. Besides the prior mentioned papers [57, 30], several other papers [87, 164] focus on predictive maintenance and real-time monitoring as among the most frequently used functionalities in construction and renovation contexts, underscoring how DTs enable a wide array of data-driven and simulation-based applications [87, 164]. For instance, Zhao et al. [164] demonstrated how a DT system, integrated with BIM, was implemented in a large hospital facility to enhance predictive maintenance by continuously monitoring equipment such as air handling units (AHU) and electrical circuits using smart meters and a diagnosis engine powered by big data and machine learning. The system enabled early detection of anomalies like low airflow and clogged filters through long- and short-term memory networks, thus streamlining maintenance schedules and reducing unplanned outages [164]. Madni et al. [87], on the other hand, offered a broader systems engineering perspective by presenting DTs as integral components in enabling condition-based and predictive maintenance through real-time monitoring, diagnostics, and prognostics. Their work emphasized the value of DTs in enhancing decision-making across the system life cycle, from design and development to operations and sustainment. Through examples from aerospace and manufacturing domains, the authors illustrated how simulation-backed DTs could identify performance degradation and optimize maintenance scheduling in advance, thereby reducing life cycle costs and increasing operational readiness [87]. This highlights the wide variety of possible applications that DT technologies offer within the construction sector, particularly in the domain of O&M. The next section will delve deeper into selected implementations of predictive maintenance to illustrate how these concepts translate into practice.

4.2 The Use Case of Predictive Maintenance

This section explores the application of DTs in the operations and facility management phase of building renovation. It outlines how DTs enhance decision-making, maintenance strategies, and overall building performance by integrating real-time data with DMs. The chapter introduces key concepts, highlights common use cases, and presents a detailed examination of predictive maintenance as a representative example, including system architectures, practical implementations, and associated benefits and challenges.

Purpose and Motivation

Predictive maintenance analyzes past operational data to assess component health and forecast when and where failures or wear are likely to occur. In contrast to *reactive maintenance* (which occurs after a failure) and *preventive maintenance* (based on fixed intervals), *predictive maintenance* optimizes intervention timing based on actual equipment behavior. This approach avoids unplanned downtime, reduces maintenance costs, and increases equipment lifespan [57].

In the context of building operations and facility management, DTs play a critical role in enabling predictive maintenance. Through its integration, DTs allow facility managers to continuously monitor the operational state of HVAC systems, elevators, lighting, and plumbing systems, detecting anomalies and planning interventions before performance degrades [30].

The goal of predictive maintenance via DTs is to enhance reliability by continuously monitoring real-time parameters, such as temperature and vibration, enabling early fault detection and timely maintenance through machine learning and virtual simulation [151].

Several motivations drive the adoption of this approach:

- **Operational Continuity:** The study by Hosamo et al. [56] emphasizes that predictive maintenance via DTs allows problems to be detected early and addressed before system failure, thereby ensuring continued HVAC performance and occupant comfort [56]. Moreover, Salzano et al. [124] demonstrate that their predictive maintenance approach significantly reduced unplanned outages and maintained uninterrupted HVAC performance at a school, with continuous post-maintenance evaluations within the DT platform confirming its long-term effectiveness in safeguarding operational continuity [124].
- **Cost Efficiency:** Hosamo et al. [56] further discuss how HVAC lifespan can be increased by at least 10%, which translates to significant cost savings [56]. Additionally, Salzano et al. [124] observed that their predictive maintenance strategy also extended the operational lifespan of HVAC components, reduced unscheduled downtime, and as a result delivered substantial operational cost savings [124]. Additionally, Mousavi et al. [98] emphasize that although initial implementation of

DTs may be costly, the reduction in maintenance frequency and downtime leads to considerable long-term cost efficiency in facility operations [98].

- **Sustainability:** The integration of DT-based predictive maintenance enables optimized asset performance, which directly contributes to sustainability goals by reducing unnecessary energy consumption and associated carbon emissions. Through comprehensive real-time perception and data-driven optimization, such systems facilitate efficient maintenance and operation strategies that minimize energy waste and support a reduced building carbon footprint [149]. Furthermore, using a model that continuously fuses high-fidelity sensor telemetry with physics-based simulations, which allows granular anomaly tracing, adaptive threshold tuning, and self-calibrating prognostics, such systems enable condition-based interventions that not only slash energy waste and unplanned downtime but also prolong equipment life cycles and drive down the overall carbon footprint of building operations [69].

System Architecture & Data Flows

In practice, a wide range of DT-based predictive maintenance architectures has emerged. While they vary in technical implementation, with some emphasizing semantic integration through ontologies, others focusing on cloud-native pipelines or edge-AI setups, they share a common goal: to unify static asset information, live sensor streams, and historical maintenance records into a coherent decision-support system for facility management. This shared foundation is evident in the following three architectural frameworks.

Cheng et al. [30] proposed a two-layer predictive maintenance architecture designed to improve how building facility managers monitor, assess, and plan maintenance for mechanical, electrical, and plumbing (MEP) systems. The framework combines BIM, IoT sensor networks, and machine learning into a seamless decision-making system for O&M. At its foundation is the information layer, which gathers data from three key sources consisting of BIM models for static asset data (e.g., type, dimensions, installation date), IoT sensors for real-time data like temperature, pressure, and flow rate, and facility management systems for historical records like maintenance logs and inspection results. These data streams are integrated using standard protocols to ensure all operational and spatial data remains connected and accessible. A plug-in then allows real-time sensor values to be visualized directly in the BIM model, providing facility managers with a live view of equipment performance.

The application layer builds on this integrated dataset with four functional modules. (1) Condition monitoring and fault alarming, which tracks sensor data for deviations and sends alerts when thresholds are breached. (2) Condition assessment, which helps classify equipment health using predefined condition scales. (3) Condition prediction, which uses machine learning models to forecast future equipment states based on historical and real-time data. (4) Maintenance planning, for scheduled repairs or part replacements before failures occur, based on predicted degradation trends.

The architecture transforms reactive and scheduled maintenance into a predictive, data-driven process. It gives building managers the tools to intervene before performance drops or breakdowns occur, improving energy efficiency, safety, and cost-effectiveness in modern facility operations [30].

Hosamo et al. [57] on the other hand, present a three-layer DT architecture for predictive maintenance of AHUs that seamlessly fuses BIM, IoT sensor networks, facility management records, and machine-learning into a unified, decision-support environment. At its foundation lies the data-acquisition layer, which continuously harvests static asset attributes such as AHU type, dimensions, materials, and installation dates; real-time operational measurements like temperature, pressure, and flow rates, via feeds exposed through a RESTful API; and historical maintenance logs and inspection reports from the facility management system. These heterogeneous streams are harmonized through a Brick-based ontology and ingested into the BIM environment via a custom plug-in, enabling facility managers to overlay live sensor readings directly onto the 3D model.

Building on this integrated dataset, the analytics layer delivers three core capabilities. First, a rule-based fault-detection engine continuously evaluates sensor patterns against expert-defined thresholds, instantly flagging anomalies and issuing alarms. Second, condition-prediction modules that employ artificial neural networks, support vector machines, and decision-tree algorithms learn from both historical records and streaming data to forecast impending component degradation. Third, a dynamic maintenance planner ingests these prognoses to prioritize and schedule interventions before failures materialize, thus shifting maintenance from a reactive or calendar-driven chore to a truly proactive, data-driven discipline.

In the final visualization and decision-support layer, all alerts, health assessments, prognostic outputs, and scheduled tasks converge within the BIM interface. By simply selecting an AHU in the Revit model, managers can review its current operational performance, examine forecasted health trajectories, and access an optimized service plan-transforming building operations into an intelligently orchestrated workflow that elevates energy efficiency, occupant comfort, and life cycle cost effectiveness [57].

Lu et al. [85] propose a five-layer DT framework to bring together all the information needed for smarter building maintenance and operations. At the bottom is the data acquisition layer, where three types of data are gathered:

1. Static details about each piece of equipment (its type, size, and installation date) from BIM exports;
2. Live measurements (temperature, humidity, vibration, energy use) from a network of wireless sensors;
3. Past maintenance records and space-usage schedules from the campus's management systems.

Next, the transmission layer makes sure the data reaches the central system reliably: sensors send minute-by-minute updates over a radio network to gateways, which forward them to a data manager; building-management controllers upload hourly CSV snapshots; and QR codes on hundreds of assets let technicians scan and log work orders via a mobile app, all funneled into the same backend. On top of that, the digital modeling layer builds a three-dimensional picture of the campus modeled at three levels of detail so that every device can be located and visualized in context.

In the integration layer, all streams like sensor readings, BIM objects, and maintenance logs come together in a NoSQL database. Each BIM object is linked to its real-world counterpart, so a single query can pull up its status, history, and spatial information at once. Two DT implementations (one research prototype and one commercial version) both tap into this shared store to stay perfectly in sync.

Finally, the service layer turns these unified data into practical tools: live dashboards and 3D views highlight any abnormal readings; built-in analytics assess current equipment health; machine-learning models forecast when parts will wear out; and maintenance schedules are automatically generated to prevent breakdowns. By organizing data and workflows in these five layers, the system shifts maintenance from reaction to prediction and makes it easy to add new sensors or analytics without overhauling the entire setup [85].

All three DT frameworks share a common foundation: they bring together static BIM-derived asset information, live IoT sensor streams, and historical maintenance records to create a unified decision-support environment for facility operations and predictive maintenance. Each leverages BIM for spatial context, overlays real-time performance data directly onto 3D models, and employs analytics to assess equipment health and forecast degradation. They all culminate in a user-facing interface that lets facility managers visualize current conditions, review trends, and schedule interventions before failures occur. What differs is how they slice this process into layers: Cheng et al. [30]’s two-layer design keeps aggregation and application tightly coupled; Hosamo et al. [57] insert a distinct analytics tier between data capture and visualization; and Lu et al. [85]’s five-layer approach further isolates data transport and 3D modeling to maximize modularity and scalability. Nonetheless, at their core, every framework is built to turn disparate data streams into actionable insights that shift maintenance from reactive to predictive.

Implementation Examples

Hosamo et al. [57] developed a predictive maintenance system tailored for air handling units (AHU) using a DT framework that integrates BIM, IoT sensors, semantic data models, and machine learning algorithms. The system collects real-time data on temperature, pressure, valve positions, and equipment status from AHU components and links it with historical maintenance records and component specifications stored in the BIM model. These inputs feed into a three-layer framework: fault detection using APAR (AHU Performance Assessment Rules), condition prediction using classifiers such

as artificial neural networks (ANN), support vector machines, and decision trees, and dynamic maintenance planning based on the predicted health status of each unit.

To validate the framework, the authors implemented it in the I4Helse building at the University of Agder, Norway. Four AHUs serving classrooms, offices, and laboratories were monitored between 2019 and 2021. A total of 25 different faults were detected using APAR rules, including simultaneous heating and cooling, incorrect damper positioning, and valve misbehavior. Machine learning models were trained on this fault data to predict upcoming failures. The ANN model achieved the highest accuracy, correctly classifying all faults and enabling maintenance forecasts up to 4.5 months in advance. These forecasts allowed facility managers to plan maintenance activities preemptively, minimizing energy waste and reducing system downtime.

As a precedent, the Sydney Opera House is referenced for its use of BIM and Industry Foundation Classes standards to centralize facilities management data. Hosamo et al. [57] extend this concept by embedding real-time sensor feedback and predictive logic, moving from static documentation toward proactive and intelligent maintenance systems. The study highlights that DTs, when combined with standardized data models and predictive analytics, can transform traditional O&M practices into scalable, energy-efficient, and preventive maintenance strategies in the built environment [57].

Bouabdallaoui et al. [24] on the other hand, aimed to create a general, scalable system for predicting HVAC equipment failures in buildings using machine learning without relying on complex or expensive setups. The project began with a sports facility in France, where a selection of HVAC units (boilers, AHU, and pumps) were fitted with IoT devices to monitor real-time data such as vibration, energy use, and surface temperature.

The collected data were streamed every hour into a cloud database. Rather than labeling each data point as “fault” or “normal”, which is often impractical in real-world buildings, the authors used an unsupervised learning approach. Specifically, they built an autoencoder model using Long Short-Term Memory (LSTM) layers [24]. LSTM units are a special kind of recurrent neural network cell designed to capture both short- and long-range dependencies in time-series signals [23]. The model then learns normal behavior patterns from the historical sensor data. If the system notices a new pattern that doesn’t match what it has seen before, it raises a fault alert.

To make this work reliably, the model calculates how different the input and reconstructed output are. If this difference, the so-called anomaly score, exceeds a threshold, a warning is triggered. The system is integrated with the facility’s Computerized Maintenance Management System, and alerts are sent to the facility manager with basic information: the faulty equipment’s name, location, and timestamp.

When tested over 45 days, the system successfully predicted two faults two days in advance, missed one failure, and issued two false alarms. Though not perfect, this early detection gave the facility managers valuable time to intervene – proving the potential of combining low-cost IoT sensors and ML algorithms for predictive HVAC maintenance.

The framework is flexible and can be retrained or transferred to other buildings and installations, adapting to their unique data [24].

Chatsuwan et al. [29] set out to give building managers an easy, affordable way to watch and care for their offices in real time, without any heavy software or coding skills. They started by taking their 3D building design and turning it into a simple web map, so one can explore each floor and room right in one's browser.

Then they hooked up the usual sensor reports for things like temperature, humidity, carbon dioxide, and occupancy counts. Instead of waiting for spreadsheets, those reports are automatically streamed into the web map every few minutes via a drag-and-drop workflow tool, so each room on the dashboard shows its latest readings and lets you click through to see how conditions have been changing.

All of that data also gets saved in a secure cloud database, encrypted in transit and authenticated at the device level, so managers can look back over days or weeks to spot slow trends, like rising CO₂ in a meeting room or air filters that need cleaning, before anyone notices a problem. This historical record supports predictive maintenance, allowing staff to schedule filter replacements, ventilation checks, or HVAC tune-ups based on data-driven forecasts rather than reactive fixes.

Because the framework is built exclusively on open-source technologies and standardized data formats, it imposes minimal technical barriers and capital expenditure. The deployment in a six-story office building in Kanagawa validated its capability to deliver both real-time environmental monitoring and longitudinal trend analysis, thereby enabling maintenance actions to be scheduled preemptively rather than reactively. By integrating continuous sensor streaming with predictive analytics, this study contributes a vendor-agnostic methodology for the rapid instantiation of DTs in facility management contexts and lays the groundwork for future investigations into automated control mechanisms and advanced maintenance strategies to further optimize building performance and occupant comfort [29].

Benefits & Challenges

In renovation and construction contexts, the implementation of DTs for predictive maintenance is challenged by limited access to original manufacturing data. Falekas and Karlis [46] note that in many cases – especially during retrofitting – engineers must rely on reverse engineering and additional testing to gather essential system information. This increases the time and effort required to develop accurate DT models. The authors emphasize that for DT use to be productive in such settings, the extra time spent on building and validating the DM must be justified within the tight operational windows typical of renovation projects. Furthermore, the integration of heterogeneous data from sensors, manuals, and models requires careful handling, as effective condition monitoring depends on high-quality, timely information across the system life cycle [46].

While DTs offer a transformative framework for operational control, their potential to support predictive maintenance depends on the effective use of performance data, real-

time monitoring, and AI-driven diagnostics. Salem and Dragomir [123] indicate that these capabilities are critical in construction settings, where identifying and addressing emerging issues proactively, such as equipment inefficiencies or environmental deviations, can help reduce delays, optimize resource use, and improve long-term system reliability [123].

Another aspect is that predictive maintenance relies on the continuous, secure exchange of sensor data between physical systems and DT platforms. However, as Alshammari et al. [5] highlight, legacy systems and BIM environments were not built with these cyber-physical demands in mind. As a result, cybersecurity vulnerabilities and interoperability gaps can disrupt data integrity, undermining the effectiveness of predictive analytics and increasing the risk of undetected faults or false maintenance triggers [5].

Furthermore, Zhao et al. [164] highlight that while DTs promise advantages in predictive maintenance, real-world applications have yielded mixed outcomes. Their analysis of case studies across healthcare, educational, and manufacturing facilities revealed that DT-driven maintenance did not consistently lead to better decision-making or significant cost savings. Some systems exhibited only marginal improvements or required substantial investment before benefits could materialize. Consequently, the actual value of DTs in predictive maintenance remains uncertain, particularly when implementation challenges such as data standards, interoperability, and life cycle integration are unresolved [164].

Thus, to better understand and justify DT adoption, especially in resource-constrained renovation and construction projects, it becomes essential to move from qualitative assumptions to quantitative analysis. The next chapter addresses this need by focusing on how to identify, structure, and measure the value of DTs, outlining economic benefit categories and methods for their evaluation in built environment applications.

Use Case Overview

Field	Details
Title	Predictive Maintenance via Building Digital Twin
Primary Actor	Facility Manager
Supporting Actors	Maintenance Technician, BIM Specialist (for plug-in integration), Machine-Learning Engineer (for prognostic models)
Stakeholders	Building Owner, Occupants, Senior Leadership
Scope	Digital twin of MEP systems including HVAC units (AHU), electrical circuits, plumbing and structural components
Preconditions	3D BIM model in place, IoT sensors deployed and calibrated, and historical maintenance logs and inspection reports ingested
Functionalities	• Real-time monitoring of parameters • Automatic fault and anomaly detection • Performance simulation of energy-management strategies, lighting configurations, and emergency procedures • Predictive maintenance scheduling via statistical and machine-learning models
Benefits	• Reduced unplanned downtime (operational continuity) • Extended equipment lifespan • Energy & cost savings through optimized maintenance • Supports sustainability goals via lower carbon footprint
Risks	• Interoperability gaps with legacy infrastructure • Up-front costs for sensor deployment & BIM integration • Data integration gaps between heterogeneous systems
Key Metrics	• Reduction in unplanned outages • Increase in equipment lifespan • Energy consumption reduction

Table 4.1: Fully Dressed Use Case – Predictive Maintenance enabled by DT

An adapted use case format based on Cockburn’s [32] template, outlining the roles, scope, prerequisites, functionalities, benefits, risks, and performance metrics of implementing predictive maintenance through a Building DT.



Value Identification and Quantification

Nowadays, businesses across various industries face significant challenges in clearly identifying and quantifying the value of technological innovations, despite increasing pressures for rapid innovation, product complexity, and personalized customer demands. Although digitization, especially concepts from Industry 4.0 such as DTs, is frequently highlighted as a promising solution to these issues, companies continue to encounter considerable difficulties in realizing and articulating clear benefits[133]. Even though DTs are widely recognized for their potential, organizations still struggle with its conception, adoption, and implementation. Practitioners particularly highlight the problem of defining a precise and valid value proposition, emphasizing the necessity of clearly identifying the specific contexts where DTs add value. Thus, there remains a notable gap in practical approaches for effectively describing and quantifying the value of DTs within industry, prompting the need for systematic business modeling frameworks tailored to DT applications [145].

This chapter addresses the current gap in value assessment by exploring systematic approaches to define, categorize, and evaluate the economic impact of DTs, with a focus on their application in the construction and building operations sector. It begins by discussing established economic evaluation frameworks and explains how these can be adapted to capture the specific benefits of DT implementations. In doing so, it reviews relevant methodologies, tools, and metrics drawn from both academic literature and industry practice. Additionally, the chapter presents practical examples from the building sector and other industries to illustrate how DT-driven value can be identified and quantified across various life cycle phases. By synthesizing these findings, the chapter highlights how DTs not only deliver measurable financial and operational benefits but can also serve as platforms for informed decision-making and proactive asset management.

5.1 Challenges in Value Identification

Despite increasing interest in digitalization, companies continue to encounter significant difficulties in realizing and articulating clear benefits from DTs. Trauer et al. [146] addressed the challenges associated with implementing DTs and emphasized that identifying the value of DTs is a key area where future research should place greater emphasis. Among the aspects they listed were:

- *“Identifying clear and valid value propositions associated with the DT solution”*
- *“Focus on when a DT can add which value ... it will be crucial to have a systematic approach to assess when the implementation of a DT really pays out and when it does not.”*
- *“How the data can be used to create added value for maintenance and development, while ensuring that the costs and benefits are in proportion” [146]*

Building on this perspective, Perno et al. [112] conducted a systematic literature review to identify the key enablers and barriers to implementing DTs in the process industry. In their paper, the authors emphasize that across several of the reviewed studies, the difficulty of identifying and articulating the value of DTs is a recurrent concern. This struggle is particularly evident in organizational settings where the novelty and complexity of DT technologies make it challenging to formulate clear value propositions or quantify potential returns. By highlighting this common thread, the paper underscores the need for more structured and empirical approaches to value assessment in future DT implementations [112].

An additional aspect of cost estimation and value identification remain critical challenges in the renovation context, as decisions often involve complex trade-offs among environmental, economic, and social objectives. As Jensen and Maslesa [68] highlight, building renovation decisions are frequently based on short-term economic considerations, which can lead to suboptimal outcomes in terms of long-term value. They argue that a broader understanding of value beyond the immediate costs is needed to support more sustainable renovation strategies. This includes addressing multiple value parameters such as energy performance, indoor climate, and architectural quality, which are often overlooked in conventional cost-focused approaches [68].

For instance, in Sweden’s “Million Homes Program,” urgent upgrades carry an estimated national cost of €30 - 50 billion, highlighting the financial scale of intervention. Khadra et al. [71] emphasize that energy efficiency measures, which typically account for only 20–25% of total renovation costs, often fail to justify investment through savings alone. This makes it essential to identify renovation strategies that generate broader economic value [71].

In this context, Life Cycle Cost Analysis and the Return on Investment framework emerge as vital tools. By evaluating investment, operational, and maintenance costs over

a building's lifespan, Life Cycle Cost Analysis enables stakeholders to assess long-term economic viability and compare alternatives under varying conditions such as discount rates and energy price escalations. This comprehensive approach supports more informed and sustainable decision-making in value-driven renovation planning [71, 21].

5.2 Identified Value Streams

To gain a better understanding of the value of DTs, they can be categorized based on the types of benefits they generate. Building on the insights from earlier chapters, the economic benefits of DTs are grouped into two main categories consisting of direct value streams and indirect value streams.

Direct value streams

DT technology delivers several direct economic benefits in building renovation projects, primarily through improved efficiency and resource optimization. One major value stream is energy efficiency and cost savings. As discussed in the prior chapter 4, integrating real-time sensor data with building models, DTs enable fine-tuned control of HVAC, lighting, and other systems to match occupancy patterns. This optimizes energy use and cuts utility costs. Studies report significant energy reductions – for example, one implementation [79] achieved about 17% lower energy consumption after deploying a building DT for optimized controls. In turn, lowering energy use directly reduces operating costs and even carbon emissions, helping meet sustainability goals. DTs also allow stakeholders to simulate retrofitting scenarios virtually before physical changes are made. By testing renovation options virtually, owners can identify the most cost-effective energy conservation measures and avoid expensive trial-and-error during construction [27]. This data-driven planning shortens project timelines and prevents rework, directly translating to capital cost savings [11].

Another direct value stream comes from maintenance and operational cost reduction [164]. By leveraging predictive maintenance, DTs shift building management from reactive fixes to proactive upkeep, helping avoid costly emergency repairs. Researchers highlight predictive maintenance as a major benefit of DT-enabled facility management, closely tied to maintenance cost savings [27]. In the prior mentioned paper by Hosamo et al. [57], a hospital case study, a DT with automated fault diagnostics led to over 10% fewer facility faults and repairs, resulting in substantial maintenance cost avoidance and improved stakeholder satisfaction. Industry surveys also show that continuously monitoring asset conditions via a DT helps anticipate maintenance needs, reducing operational expenses and streamlining management workflows. In summary, by eliminating inefficiencies and optimizing operations, DTs directly lower energy bills, reduce maintenance costs, and generate significant project savings in renovation efforts. These tangible economic benefits make a compelling business case for integrating DTs into retrofit and building operation strategies [27].

Indirect value streams

Beyond these immediate cost and energy impacts, DTs create indirect value streams that improve the long-term performance and sustainability of renovated buildings. One crucial indirect benefit is improved reliability and uptime of building systems. With predictive insights from a twin, facility managers can address issues before they escalate, minimizing unplanned downtime of critical equipment [149]. For example, a twin continuously monitoring a building's mechanical systems can detect anomalous behavior (such as an HVAC component deviating from normal parameters) and alert staff to intervene early. This proactive approach ensures that building services (heating, cooling, elevators, etc.) experience fewer unexpected outages, meaning less disruption to occupants and business operations. Although the monetary value of avoided downtime can be hard to quantify, it is significant – reduced system outages protect revenue in commercial buildings and maintain tenant comfort, thereby safeguarding the building's value proposition in an indirect but important way. By enabling predictive maintenance strategies, DTs help shift maintenance to off-peak times or planned intervals, which bolsters reliability and keeps facilities running smoothly [12].

DTs also contribute to extended asset lifespan. Through continuous data acquisition from sensors and simulation-based behavior modeling, DTs allow for early fault detection, enabling proactive interventions before minor defects evolve into catastrophic failures that could shorten equipment lifespan [76]. In the mining industry specifically, DTs have been integrated with AI-driven monitoring systems to detect anomalies under extreme operational conditions, which has demonstrably led to longer equipment life cycles and fewer unexpected downtimes. This proactive asset management approach directly supports the deferral of capital expenditures by reducing the frequency with which large-scale replacements [120]. While exact figures in building renovations vary, the trend is clear: leveraging a DT for smart maintenance translates into assets that remain serviceable for longer with lower total cost of ownership [76, 120].

Furthermore, enhanced sustainability and regulatory compliance are indirect but significant value streams enabled by DT technologies in renovation contexts. By optimizing building operations, DTs can reduce energy consumption and CO₂ emissions, thereby lowering environmental impact and operational costs [27]. DTs can support alignment with sustainability objectives and green building standards. For example, these systems allow predictive modeling of energy use patterns and occupant behavior, enabling proactive strategies for energy efficiency [129, 161]. Advanced platforms can offer continuous dashboards that display sustainability metrics, helping managers balance occupant comfort with ecological goals [152].

Moreover, DTs facilitate scenario-based analyses like simulating the effect of solar panels or improved insulation without requiring physical modifications, allowing for data-driven decision-making [107]. These simulations have the direct benefit of enabling cost-benefit assessments that inform retrofitting strategies aimed at decarbonization. While indirectly, the sustainability improvements gained through DTs enhance an organization's reputation

and align properties with future market and regulatory demands for low-carbon, resilient buildings [27, 152].

5.3 Economic Frameworks for Quantification of DT Value

As outlined in Chapter 3, several frameworks were reviewed to support the quantification of DT value. Life Cycle Cost Analysis and Return on Investment were selected as the primary evaluation methods due to their relevance and applicability. In addition, use cases from the building industry as well as those specifically related to DT applications were examined to provide a practical foundation for assessing and understanding the economic and operational benefits of DTs.

Life Cycle Cost Analysis (LCCA)

To achieve sustainable construction, it requires decision-making frameworks that account not only for economic performance but also for environmental and social impacts throughout a building's entire life cycle [6]. Life Cycle Cost Analysis (LCCA) has emerged as an economic evaluation method that considers all costs of owning, operating, maintaining, and disposing of a building or system over its entire lifespan, whilst it enables stakeholders to compare the costs and sustainability implications of alternative project designs [6, 7]. In the construction sector, LCCA is valued as a comprehensive framework for assessing “the money's worth” of a project by accounting not just for initial capital expenses but also future costs such as operation, maintenance, and end-of-life disposal. This holistic approach supports more sustainable decision-making, ensuring that short-term savings do not lead to higher long-term expenses as O&M costs are often a dominant part of a facility's life cycle cost, making LCCA currently one of the most well-known methodologies for sustainability [7, 40].

Definition

The Guidelines for LCCA from Stanford University by Davis et al. [34] define LCCA as a systematic process for evaluating the total economic impact of a building throughout its entire life span. Rather than focusing solely on initial capital expenses such as design and construction, LCCA encompasses all future costs including utilities, operations, maintenance, repairs, and eventual replacements. This comprehensive approach, sometimes called “whole cost accounting” or “total cost of ownership,” allows project teams to make more informed design decisions by analyzing how different building systems and configurations affect both short-term expenditures and long-term value [34].

Implemented within Stanford's Project Delivery Process (PDP) [137], which is a standardized framework for guiding capital projects through all phases, LCCA requires the development of benchmark budgets using historical cost data, followed by comparative analyses of alternative design options across various building systems-such as HVAC, lighting, glazing, and structural components [34, 137]. Each alternative is assessed using

key financial metrics, notably net present value and payback period, to determine cost-effectiveness as done in [13, 95]. For example, alternatives with a payback period of five years or less must be adopted, while those with paybacks of six to ten years are strongly encouraged unless exempted by senior administration. These evaluations incorporate the time value of money through discount and escalation rates, and while many assumptions are necessary, they are consistently documented to ensure transparency and consistency. Ultimately, the LCCA framework equips project teams with the tools to balance initial investments against long-term performance, encouraging the selection of systems that offer both cost-effectiveness and, often, enhanced sustainability [34].

LCCA in the Building Industry

For the conduction of the LCCA analysis, information across the various life cycle phases of a building is required. Shin and Cho [134] differentiated between the construction, operation, and disposal phases. Specifically for the operation phase, two key cost categories are considered: maintenance, repair, and replacement costs, as well as operation energy costs. The present worth of maintenance, repair, and replacement costs is calculated by multiplying the quantity of each item requiring such actions by its respective unit cost, and discounting these values over time using an appropriate discount rate. Similarly, the operation energy cost is determined based on the annual energy consumption and the unit cost of energy, which is also converted into present value through discounting over the operation period. Therefore, to conduct a comprehensive LCCA during the operational phase, it is essential to obtain detailed information on quantities, emission factors, and additional parameters such as repair cycles, replacement rates, energy usage, and energy prices.

To then conduct the LCCA, a framework was developed using data mapped from BIM to the required LCCA inputs. BIM facilitates the extraction of key data such as material quantities and annual energy consumption, which are essential for calculating costs in the construction, operation, and disposal phases. The LCCA framework was implemented in an Excel-based tool, with separate worksheets for each phase and an automated summary sheet. In the operation phase, LCCA includes energy costs and maintenance, repair, and replacement costs, which are calculated using user inputs like energy prices, unit costs, and operation cycles, all of which can be derived from or supported by BIM data [134].

Building upon this methodology, Rostamiasl and Jrade [122] proposed an enhanced process for integrating BIM with LCCA specifically aimed at assessing the economic viability of aging-in-place residential designs. Their approach leveraged Autodesk Revit to extract and tag O&M-relevant building elements (e.g., plumbing systems, finishes, appliances) and assign life cycle attributes, such as expected time to replacement or maintenance frequency. These attributes were input into a structured spreadsheet-based LCCA environment capable of modeling different O&M scenarios. Importantly, the authors demonstrated how scenario simulations, such as increasing HVAC maintenance frequency or altering energy tariffs, can significantly impact total operational costs over a 30-year period. This underscores the value of BIM-integrated LCCA not only in providing

cost transparency but also in enabling data-driven, predictive decision-making throughout the O&M phase [122].

In a comprehensive review, Lu et al. [84] examined the state of BIM-based LCCA. Their study aimed to clarify the current methodologies, tools, and applications used in integrating BIM with LCCA. They identified common processes involving data input, cost calculation, and output visualization across different life cycle phases and categorized various data exchange approaches-such as exporting from BIM to spreadsheets, linking to external software, or embedding LCCA directly within the BIM environment – and highlighted their respective advantages and limitations. For example, Santos et al. [125] developed a process using the industry foundation classes (IFC) schema, which is an often-used information exchange format, to map Life Cycle Costing (LCC) related data, while Jalaei et al. [66] integrated cost information into the BIM tool Revit through API scripting for more automated LCCA. Additionally, Kehily and Underwood [70] embedded LCC calculations directly within a 5D BIM environment. This shows the wide variety of opportunities for advancing the use of BIM in enhancing the accuracy, efficiency, and strategic value of LCCA in building projects [84].

LCCA DT Application

Whilst DT technology is increasingly integrated into Life Cycle Cost Analyses (LCCA), its own value remains largely unquantified as research mainly focuses on using DTs to enhance LCCA, rather than applying LCCA to measure the value delivered by the twins themselves[41]. In the paper by Borjigin et al. [19] on DTs for light rail infrastructures, LCCA was applied alongside Life Cycle Assessment (LCA) within a BIM-based environment to holistically assess both environmental and economic impacts across the infrastructure life cycle. The authors implemented LCCA to convert costs from various project phases into present values, thereby enabling better-informed decisions during early-stage design. The study found that DT integration enabled parametric sensitivity testing (e.g., varying grout volumes), which uncovered cost-intensive components such as unrecyclable bitumen and grout. Importantly, the authors emphasized that such insights allow for real-time design adjustments in DT environments, which significantly enhances the sustainable value of infrastructure investments. This substantiates their claim that DT-aided LCCA facilitates proactive cost optimization and supports sustainable development decisions throughout the project life cycle [19].

González-Prida et al. [51] extend LCCA into the energy systems domain by focusing on power converters. Their approach integrates digitalization and an Asset Health Index (AHI) into a strategic LCCA framework. This integration allows dynamic monitoring of asset degradation and cost implications over time, enabling more precise predictive maintenance scheduling. Their nine-step methodology includes acquisition, operation, maintenance, and disposal costs. The key contribution lies in coupling DT-generated data with AHI to inform recurring maintenance and investment decisions. They conclude that this data-driven, condition-based LCCA not only optimizes asset reliability

but also supports sustainability through cost-effective life cycle planning and resource allocation [51].

Lastly, Serugga [128] explores how DTs and AI models can enhance cost modeling in off-site modular construction. His research incorporates LCCA by using DTs for real-time tracking of component behaviors and cost-performance metrics throughout the OSC life cycle. The model supports predictive analytics for automated material take-offs and life cycle cost estimation, especially addressing the high variability typical of modular systems. Importantly, Serugga [128] argues that integrating DTs into LCCA allows for simultaneous management of customization demands and modular efficiency, leading to a new level of adaptive, cost-conscious design and delivery. The paper validates this via case studies that demonstrate the model's ability to reduce cost variability while improving decision agility across the construction life cycle [128].

Return on Investment

Definition

Return on Investment (ROI) is a widely used metric that helps evaluate the efficiency and effectiveness of an investment by comparing the financial return to the associated costs. Fundamentally, ROI is defined as the ratio between the net gain from an investment and its cost, often expressed as a percentage, providing a standardized way to assess profitability across different initiatives and sectors [21]. It originated as a financial tool grounded in rigorous accounting principles and has since become a ubiquitous metric in both the private and public sectors due to its simplicity, clarity, and ease of interpretation [135].

The basic formula for ROI is:

$$\text{ROI} = \left(\frac{R - C}{C} \right) \times 100$$

Where the net benefit typically refers to the gain achieved from the investment after subtracting all relevant costs [21]. Despite its conceptual straightforwardness, ROI calculations can vary depending on context, which has led to its use as an umbrella term encompassing a variety of financial evaluations, such as benefit-cost ratios, internal rate of return (IRR), and payback periods [135].

The popularity of ROI stems from both objective and subjective factors. Objectively, it promotes cost-efficiency, allows comparison across diverse projects, and supports accountability through transparent use of financial data. Subjectively, ROI feels intuitive due to its presence in academic training and personal finance, offering a single numerical value that simplifies decision-making [21].

However, its limitations are equally recognized. ROI has been criticized for oversimplifying complex decision-making processes and potentially misrepresenting long-term value by focusing narrowly on quantifiable financial outcomes. For instance, Jacobson [64]

points out that ROI, as typically measured through accounting data (e.g., net income over total assets), may fail to accurately reflect true economic performance, especially when misaligned with stock return data, which is a more market-sensitive performance indicator [64]. In modern contexts, ROI is increasingly applied beyond purely financial returns to encompass broader organizational impacts, including quality improvement, cost avoidance, and productivity enhancements, as seen in healthcare quality frameworks [132]. This evolving usage highlights its versatility but also underscores the need for careful and consistent application to preserve its analytical value.

ROI in the Building Industry

The assessment of ROI for BIM requires detailed evaluation of cost-saving impacts across various phases of a building's life cycle. Similar to LCCA, ROI must consider both direct and indirect benefits realized through BIM adoption. Giel and Issa [50] propose a structured ROI estimation based on comparative case studies that quantify cost differences between BIM-assisted and non-BIM-assisted projects. These studies specifically focus on measurable benefits such as reductions in Requests for Information (RFIs), change orders, and schedule overruns.

In construction phases, direct cost savings attributable to BIM include fewer design conflicts and enhanced coordination, leading to measurable reductions in change orders and RFIs. For example, case studies demonstrated reductions in RFIs by 34–68% and in change orders by 37–48%, depending on project type and complexity. The ROI values observed ranged dramatically – from a modest 16% to an exceptional 1,654% – highlighting the variability based on project scale and BIM implementation timing [50].

From an operations and maintenance perspective, ROI calculations must account for long-term savings enabled by enhanced facility management practices. BIM provides a digital, data-rich model that facilitates proactive maintenance and asset tracking. Gao and Pishdad-Bozorgi [49] emphasize that despite BIM's acknowledged potential in O&M since 2010, its adoption remains low due to poor interoperability and lack of alignment between BIM data and facility management systems. Nevertheless, when implemented, BIM significantly aids in reducing emergency maintenance needs and improving energy performance tracking, both of which are critical to LCA cost optimization [49].

Abideen et al. [2] highlight that the O&M phase constitutes approximately 60% of total life cycle costs, underscoring the importance of assessing ROI within this phase. The study identifies that BIM-integrated O&M systems can lead to substantial cost savings through functions such as maintenance scheduling, asset information retrieval, and energy monitoring. Moreover, inadequate integration of BIM with Computerized Maintenance Management Systems is a major barrier; yet when integration is successful, ROI improves through reductions in reactive maintenance and better life cycle asset management [2].

Operational ROI is further enhanced by the digitization of maintenance logs, predictive maintenance planning, and streamlined access to building data, all of which minimize labor costs and extend asset life. BIM-enabled O&M also supports visualization and

simulation of facility systems, allowing facility managers to simulate various maintenance scenarios and budget forecasts [2].

Walasek and Barszcz [154] advocate for the development of better interoperability standards to address ROI realization during O&M. They note that although many BIM applications demonstrate potential ROI during construction, future research must aim at improving measurement frameworks for the O&M stage [154].

In conclusion, calculating ROI from BIM should involve:

- **Quantify reductions in RFIs, change orders, and schedule delays during design and construction** [154, 50]
- **Estimate long-term savings from energy management and predictive maintenance** [49, 50]
- **Consider interoperability barriers and user-specific requirements in O&M phases** [49, 50]

Despite implementation challenges, the reviewed studies consistently support BIM as a financially viable tool, particularly when O&M considerations are integrated from the outset.

ROI DT Application

The assessment of ROI in DT applications varies across different sectors and is influenced by context-specific financial, operational, and technological considerations, as shown in the following section. In the manufacturing sector, Malik [89] proposes a structured, simulation-based framework that explicitly connects technological investments in DTs to financial outcomes. His model segments DT users into four categories, developers, integrators, commissioners, and users, each with distinct investment profiles and ROI considerations. The methodology evaluates life cycle phases such as design, commissioning, and maintenance through detailed use cases. Malik emphasizes that while upfront development costs may be high, long-term operational savings and process improvements justify these investments, provided that the ROI calculation accounts for the complexity and scale of the DT implementation [89].

Similarly, Banyai and Kovács [14] examine DTs in the context of job-shop manufacturing. Their research introduces a simulation-supported methodology that links productivity metrics such as lead time and capacity utilization to financial indicators. Their model incorporates agent-based simulations to evaluate the comparative ROI of different identification technologies. This study illustrates the value of technology-specific ROI analysis, demonstrating how DT adoption strategies must be tailored to both the technological context and the financial constraints of the enterprise[14]. In a different industrial context, Deshpande and Deshpande [38] explore the application of DTs in the printing and

packaging industry, where dynamic production schedules, custom orders, and regulatory compliance drive digital innovation. DTs here are primarily leveraged to optimize supply chain logistics, increase Overall Equipment Effectiveness, and support remote diagnostics and predictive maintenance. Measurable ROI is observed in multiple areas: printing equipment manufacturers using DTs report up to a 15% increase in customer productivity and 50% reduction in training costs, while support service efficiency rose by 70%. These financial impacts are enabled by virtualized fault diagnosis, real-time performance tracking, and centralized job management, all characteristics of an effective DT system [38].

In the renewable energy sector, Cao et al. [26] assess the economic outcomes of DT-driven enterprise digital transformation. Based on a global empirical study of over 200 companies, they find that DT adoption in investment decision-making yields, on average, a 15% increase in ROI and a 20% reduction in decision cycle time. DTs enable companies to better forecast energy consumption, optimize resource allocation, and reduce risk, which collectively improve the economic viability of green energy projects. The study demonstrates that the strategic use of DTs, when embedded into enterprise-wide decision-making frameworks, significantly enhances both speed and accuracy of investment decisions [26].

Also in high-tech and capital-intensive domains like aerospace and defense, Malone [90] emphasizes that ROI estimation for DTs is deeply influenced by model fidelity, development effort, and system complexity. His research highlights the application of the U.S. Department of Defense's Better Buying Power [35] strategy, where ROI is framed not just as a financial figure but as a function of increased capability, reduced acquisition time, and enhanced life cycle performance. For example, in mission-critical simulations, the presence of validated DTs enables earlier failure detection, supports virtual testing, and reduces operational costs through predictive maintenance. However, it is cautioned that traditional ROI calculations often overlook the scalability and integration costs of DT systems, especially in multistakeholder environments like defense procurement [90].

The McKinsey & Company [92] article further validates the importance of DTs in optimizing public infrastructure investment. Their findings suggest that DTs enable governments to virtually test and refine infrastructure projects before physical implementation, reducing costly design errors, accelerating timelines, and improving asset utilization. As a result, they advocate for the integration of DTs as a tool for not only project delivery but also ROI evaluation, particularly in large-scale public infrastructure portfolios where transparency, accountability, and long-term performance are paramount [92].

A key contribution to this discourse is the work by Caccamo et al. [25], who position the DT as both a subject of ROI analysis and a tool to conduct it. Importantly, they propose that a DT need not be implemented across an entire production line; even partial or cascading DT implementations targeting critical components can yield meaningful ROI. By simulating expected outcomes before making real-world investments, DTs offer a low-risk, high-fidelity mechanism for financial forecasting and operational improvement, particularly valuable in dynamic, high-mix, low-volume manufacturing settings [25].

In conclusion, the reviewed literature shows that while ROI evaluation in DTs is context-dependent and often methodologically complex, it is increasingly supported by simulation, life cycle integration, and multi-scenario analysis. DTs not only enhance operational and financial performance but also act as platforms for validating investment decisions before physical execution. Thus, DTs are not merely technologies to be evaluated for ROI – but powerful evaluative tools.

Conclusion

6.1 Summary and key findings

In this thesis, the two main analytical chapters have laid the foundation for understanding how DTs can transform building renovation projects and how their economic impact can be measured. The following revisits the main research questions, summarizing how each was answered based on the findings in Chapters 4 and 5.

“What are the primary use cases for Digital Twin technology in building renovation projects?”

Chapter 4 addressed this question by identifying key applications of DTs across the renovation life cycle. In the planning phase, DT-driven simulations enable rapid comparison of energy-efficiency strategies and assessment of emergency scenarios. During condition assessment, the integration of IoT sensor streams into a live DM facilitates early anomaly detection, flagging HVAC faults or structural irregularities before they escalate. The use case of predictive maintenance during the operations and facility management phase was highlighted as particularly important, as it supports operational continuity, cost efficiency, and sustainability. More specifically, by combining BIM-based asset data, IoT sensor feeds, and historical maintenance records, DTs enable continuous condition monitoring, early fault detection, and optimized intervention scheduling. This shift from reactive or time-based maintenance to proactive, data-driven strategies reduces downtime, extends equipment life, lowers operational costs, and supports sustainability goals by improving energy efficiency and reducing carbon emissions.

To enable these applications, DT frameworks typically integrate static asset information, real-time data streams, and historical records within unified decision-support environments. Architectural implementations range from simple two-layer setups to more modular five-layer designs, varying in complexity but sharing the goal of delivering reliable condition assessment, fault prediction, and maintenance planning. Collectively,

these use cases demonstrate the potential of DTs to enhance asset management, operational efficiency, and lifecycle optimization, while also revealing challenges such as data interoperability, cybersecurity risks, dependence on high-quality input data, and varying Return on Investment outcomes.

“How can the value generated by Digital Twins be assessed?”

Chapter 5 explored how the earlier discussed functional benefits of DTs can be linked to measurable value. To provide structure, a two-stream value taxonomy was proposed, distinguishing between direct gains (e.g., reduced maintenance costs, lower energy consumption) and indirect gains (e.g., extended asset lifespan, improved decision-making, enhanced sustainability). This framework enables practitioners to map specific DT functionalities to both tangible and intangible value categories. However, there are many common barriers to value identification due to the novelty and complexity of DT systems, highlighting that renovation projects often need to balance multiple, and sometimes competing, objectives.

“What methods or metrics can be used to evaluate and quantify the economic value of Digital Twins in renovation projects?”

Chapter 5 furthermore examined practical methods for quantifying the identified value. Two main approaches were proposed: Life Cycle Cost Analysis and the Return on Investment framework. Firstly, Life Cycle Cost Analysis was positioned as a comprehensive method for translating benefits into present-value terms, supported by BIM-driven data extraction and scenario simulations that enable whole-life economic decision-making across construction, operation, and disposal phases. An interesting finding was that Life Cycle Cost Analysis was consistently used in combination with DTs rather than solely as a tool for value assessment, highlighting the versatility of DT applications. Secondly, Return on Investment frameworks, when adapted for DT applications, combine simulation, life cycle integration, and multi-scenario analysis to assess financial returns. Here, various studies showed that the application of DTs in the context of the building industry can provide a significant increase in the ROI assessment of DTs. In this context, calculating Return on Investment from BIM-enabled DTs should consider both short-term benefits, such as reduced RFI, change orders, and schedule delays, and long-term savings from improved energy management and predictive maintenance, while also accounting for interoperability challenges and user-specific requirements. These two methods have been applied in contexts ranging from building systems and modular construction to defense procurement and renewable energy. Overall, they both offer a structured way to assess the value DTs can create and robust tools to justify investments in renovation projects, whilst also using DTs themselves as tools for more accurate Return on Investment and Life Cycle Cost Analysis assessments.

In conclusion, this thesis shows that DTs can meaningfully improve building renovation across the whole building life cycle, particularly during the operations and maintenance phase, where predictive maintenance delivers significant reductions in operating costs and sustainability impacts. Furthermore, as value quantification becomes increasingly

important, promising approaches already exist for assessing DT economics, supporting evidence-based investment decisions and guiding future research in the coming years.

6.2 Discussion

Despite the clear promise demonstrated, practical adoption of DTs in renovation remains constrained by several interlinked challenges. First, a lack of standardized terminology and semantic data models inhibits seamless interoperability among BIM platforms, IoT systems, and facility management systems. Second, retrofitting legacy buildings with sufficient sensor density and aligning heterogeneous data streams to high-fidelity digital representations incurs significant upfront costs and requires specialized technical expertise barriers that are particularly acute for small contractors and financially constrained public agencies. Third, while simulations and retrospective analyses attest to projected benefits, longitudinal empirical studies that track DT performance and economic outcomes over multiple renovation life cycles are scarce; without such validation, risk-averse stakeholders remain hesitant to invest. Finally, organizational and governance gaps, which range from unclear roles in data management to insufficient workforce training, undermine the ability to translate technical innovations into operational routines. These barriers underscore that technology alone is not the solution and have to be overcome to achieve a robust, well-governed, and scalable DT system in renovation projects.

6.3 Future Work and Research Directions

Future work should broaden and deepen the evidence base for DTs in renovation by (1) conducting long-term, comparative studies across diverse building types and project scales, (2) refining interoperable data standards and simple integration toolkits so different platforms connect with minimal customization, (3) developing practical guidance on governance, skills, and change management to embed DT use in everyday workflows, and (4) establishing clearer, reusable economic evaluation templates to consistently capture costs and benefits. More generally, researchers and practitioners should collaborate on simple, scalable approaches, both technical and organizational, that lower entry barriers and make DT adoption an essential part of the building industry.

Overview of Generative AI Tools Used

To enhance the formulation and clarity of this thesis, I used the generative AI tool ChatGPT; all research, analysis, and substantive content were, however, conducted independently by the author.

Übersicht verwendeter Hilfsmittel

Zur Verbesserung der Formulierungen und Klarheit dieser Arbeit nutzte ich das generative KI-Tool ChatGPT; sämtliche Recherchen, Analysen und inhaltlich-substantiven Beiträge wurden jedoch eigenständig von der Autorin erbracht.

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Acronyms

AEC-FM Architecture, Engineering, Construction, and Facility Management.

AHI Asset Health Index.

AHU Air Handling Units.

AI Artificial Intelligence.

AR/VR Augmented/Virtual Reality.

BIM Building Information Modeling.

CPS Cyber-Physical Systems.

DM Digital Model.

DS Digital Shadow.

DT Digital Twin.

EOL End-of-Life.

EU European Union.

HVAC Heating, Ventilation, and Air Conditioning.

IFC Industry Foundation Classes.

IoT Internet of Things.

IRR Internal Rate of Return.

LCA Life Cycle Assessment.

LCC Life Cycle Costing.

LCCA Life Cycle Cost Analysis.

LCE Life Cycle Energy.

LSTM Long Short-Term Memory Layers.

MEP Mechanical, Electrical, and Plumbing Systems.

NASA National Aeronautics and Space Administration.

O&M Operation and Maintenance.

PLM Product Life Cycle Management.

RFI Requests for Information.

RFID Radio Frequency Identification.

ROI Return on Investment.

SME Society of Manufacturing Engineers.

UWB Ultra-Wideband.

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